

When Does Defense Randomization Defeat FL Poisoning? A Persistence Boundary

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Abstract

Server-side per-round defense randomization is widely assumed to provide security value against FL poisoning by exploiting the adversary’s uncertainty about which aggregation rule will be sampled. **Our central finding is negative:** across the FL deployments we test (CIFAR-10/100, model_scaling/backdoor_pixel/DBA/edge_case attack families, NormClip/reputation/trimmed_mean defense menus), realized value of private defense (VoPD) is statistically indistinguishable from zero or only marginally positive (-0.010 to $+0.032$ across $n = 30$ seeds). The mechanism is that *persistent* attacks make temporal mixtures max-like rather than average-like: once a backdoor embeds, intermittently sampling a suppressing defense recovers no value. We support this with (a) a Markov persistence dynamic whose admissibility-conditional refinement is quantitatively predictive on high-admission policies (out-of-sample MAE 0.041–0.059); (b) a checkable $O(|\mathcal{A}||\mathcal{S}|)$ static diagnostic (Theorem 1, Howard 1966 VOI tightness instantiated for empirical FL); (c) an empirical boundary map with *two explicit existence proofs* of survival — the stateless spam game ($\gamma = 0$, VoPD 0.200) and a restricted reputation+trimmed_mean menu in FL whose natural Nash equilibrium places reputation at 21.8% with realized VoPD $+0.023 \pm 0.039$ at $n = 30$ (24/30 positive). These existence proofs are not deployable policy recommendations; the deployment implication is also negative — across realistic FL deployments, the operationally correct policy is pure FedAvg, and the diagnostic’s value is to certify that without requiring deployers to build randomization infrastructure speculatively.

1 Introduction

Federated learning (McMahan et al., 2017) enables collaborative model training without centralizing data, but its distributed nature introduces vulnerability to data poisoning attacks (Chen et al., 2017; Bagdasaryan et al., 2020). An escalating arms race between attacks and defenses (Wang et al., 2020; Xie et al., 2020; Pillutla et al., 2022; Sun et al., 2019; Bhagoji et al., 2019) has produced strong results on both sides, but these analyses share two key assumptions: (1) the adversary knows the exact defense employed, and (2) the adversary has access to samples from the true data distribution. In practice, the server’s defense is an implementation detail that can and should be kept private, and the adversary’s optimal play depends critically on which defense is active.

Moreover, defense choices are not free: each mechanism imposes accuracy and fairness costs (Wang et al., 2020), and attack strength depends on network heterogeneity, the defense in use, and model architecture. A game-theoretic formulation naturally captures these mutual dependencies.

Contributions.

- 1. Persistence mechanism + admissibility-conditional quantitative refinement (most novel contribution).** Persistent attack state makes temporal mixtures *max-like* rather than average-like: once a backdoor embeds, intermittently sampling a suppressing defense recovers no value. We formalise this as a Markov persistence dynamic ($\gamma, \alpha, p_{\text{eff}}$); the qualitative heuristic (Proposition 4.2) is $\sim 20\times$ slack, but conditioned on a per-seed admissibility latent the heuristic becomes quantitatively predictive *in the high-admission* ($p_{\text{admit}} \geq 0.83$) regime: out-of-sample MAE = 0.041 on pure NormClip and 0.059 on the NE3 mix (§4.2, Appendix K.1). *Pure FedAvg* ($p_{\text{admit}} = 0.50$) remains qualitative-only (MAE = 0.426, dominated by Bernoulli noise). The mechanism is empirically validated by the $\tau=2$ disentanglement (§5.4, Appendix H, $n = 15$): lowering admission via stricter clipping does not reduce realized scaling ASR (0.955 at $\tau = 2$ vs. 0.961 at $\tau = 5$, indistinguishable).

2. **Negative result + boundary characterisation: in realistic FL, randomization essentially does not help (the scientific contribution).** Across the range we tested — CIFAR-10/100, four persistent attack families, multiple defense menus — realized VoPD is statistically indistinguishable from zero or marginally positive ($[-0.010, +0.032]$). On all menus where the natural NE under standard cost weights is reachable on the 4-defense menu, persistence collapses the static VoPD prediction. *Two existence proofs* show the boundary is real, not vacuous: the stateless spam game ($\gamma = 0$, realized VoPD = 0.200, δ_{NE} recovered); and a restricted reputation+trimmed_mean menu in FL where the natural 2-defense NE places reputation at 21.8% with realized VoPD $+0.023 \pm 0.039$ at $n = 30$ (24/30 positive; §5.9). *These are existence proofs, not deployable policy recommendations.* The boundary is determined by the defense menu’s arg max structure (does any single attack dominate all menu members?), not by attack-menu cardinality or architecture (Lemma 1, §5.8–5.9). Detailed boundary map: (a) on **NormClip+reputation**, realized VoPD = $+0.002 \pm 0.031$ at the static-NE mix — the persistence collapse is robust (§5.5); (b) the mix-ratio sweep (§5.6) maps the collapse boundary within this menu at reputation weight $\sim 70\text{--}90\%$ (scaling ASR drops from 0.945 at 10% reputation to 0.097 at 90%); (c) on **reputation+trimmed_mean** at the rep30/tm70 mix (§5.8), a per-round-switching oracle adversary at $n = 30$ achieves realized full-info ASR 0.915 ± 0.065 vs. best-committed 0.871 ± 0.095 (scaling) or 0.747 ± 0.106 (pixel) — **realized VoPD = $+0.032 \pm 0.054$ (27/30 seeds positive); complementarity survives at a modest magnitude;** (d) the survivor regime is robust against a structurally distinct third persistent attack family (backdoor_edge_case, §5.10): realized 4-policy VoPD = $+0.060 \pm 0.013$ (5/5 seeds positive, 4/5 STRONG band), *reproducing the survivor result with the third attack absorbed*; (e) architecture replication on ResNet18 + GroupNorm (§5.11) preserves the qualitative monotonic ordering but the dramatic NC10/rep90 suppression does not replicate (0.865 on ResNet18 vs. 0.097 on cifar_cnn) — *the collapse boundary’s quantitative location is architecture-conditional*, while its qualitative existence holds; (f) **NE-reachable survivor on the restricted {reputation, trimmed_mean} menu** (§5.9): the natural 2-defense Nash equilibrium places reputation at 21.8% and trimmed_mean at 78.2% with static VoPD = **0.180**; the realized 30-seed BR at this NE point gives realized VoPD = **$+0.023 \pm 0.039$** (24/30 positive; persistence collapses $\sim 87\%$ of the static gap but does not eliminate it) — a survivor regime that is *simultaneously deployable AND NE-selected*, answering the reviewer-flagged objection that the rep+tm survivor is only deployable-but-not-NE-reachable under the standard 4-defense menu (§5.7 λ_c sweep).
3. **Operational diagnostic (Theorem 1, Algorithm 1, supporting machinery).** A checkable $O(|\mathcal{A}||\mathcal{S}|)$ test — Howard 1966 VOI tightness instantiated for empirical FL payoff matrices — gives VoPD > 0 iff no single attack is best-response to every defense in equilibrium support. Algorithm 1 (measure-then-verify) wraps it as a deployment screening tool. The diagnostic correctly identifies positive cases (spam, $\delta_{\text{NE}} = 0.200$; the rep+tm restricted-menu NE, static VoPD 0.180) and correctly flags persistence regimes via the verify step. The operational guarantee at $k=10$ is a bootstrap-stability corollary (Proposition 2, 97.8% under representativeness); the worst-case Hoeffding bound ($k_{\text{req}} = 726$) is in Appendix L.

2 Preliminaries

2.1 Federated Learning

Consider a federated system with N clients and a central server. Each client i holds a local dataset $\mathcal{D}_i$ drawn from a possibly distinct distribution P_i . The goal is to collaboratively learn a global model w^* that minimizes:

$$w^* = \arg \min_w \sum_{i=1}^N \frac{|\mathcal{D}_i|}{|\mathcal{D}|} F_i(w), \quad F_i(w) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\ell(w; x, y)] \quad (1)$$

In FedAvg (McMahan et al., 2017), each round t the server broadcasts w^t , selected clients perform local SGD, and the server aggregates: $w^{t+1} = w^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$.

2.2 Data Poisoning in Federated Learning

An adversary controls a fraction f of clients. These malicious clients can:

- **Label flipping:** Change labels of training samples (Chen et al., 2017).

- **Backdoor injection:** Insert triggers to induce targeted misclassification (Bagdasaryan et al., 2020; Xie et al., 2020).
- **Model scaling:** Amplify malicious updates to dominate aggregation (Bagdasaryan et al., 2020).

2.3 Byzantine-Robust Aggregation

To counter poisoning, the server may replace FedAvg with robust aggregation rules:

- **Krum** (Blanchard et al., 2017): Selects the update closest to its neighbors.
- **Trimmed Mean:** Discards extreme values coordinate-wise before averaging.
- **Coordinate-wise Median:** Takes the median along each coordinate.
- **Norm Clipping** (Sun et al., 2019): Clips updates exceeding a threshold τ .
- **RFA** (Pillutla et al., 2022): Computes the geometric median of updates.

2.4 Game Theory Basics

A normal-form game is a tuple $G = (\mathcal{N}, \{\mathcal{S}_i\}_{i \in \mathcal{N}}, \{u_i\}_{i \in \mathcal{N}})$ where $\mathcal{N}$ is the set of players, $\mathcal{S}_i$ is the strategy space of player i , and $u_i : \prod_j \mathcal{S}_j \rightarrow \mathbb{R}$ is the payoff function.

A *Bayesian game* extends this with a type space Θ_D and prior belief $\mu \in \Delta(\Theta_D)$. A strategy profile forms a *Bayes-Nash Equilibrium (BNE)* if each player maximizes expected utility given their beliefs about others' types.

3 Game-Theoretic Framework

3.1 System Model

We consider an FL system with N clients where a fraction f are adversarial. The server employs a defense mechanism from a finite set, and critically, *the choice of defense is private*—the adversary does not observe it.

Remark (Randomization vs. Secrecy). The defense *mechanism* is public; what remains private is the *realized choice* each round. The VoPD derives from this per-round secrecy, not from hiding the algorithm class (Kerckhoffs's principle). See also the BNE vs. Stackelberg discussion in §3 and Appendix C.

3.2 Players and Strategy Spaces

Player 1 (Adversary, A): Controls fN malicious clients. Strategy space:

$$\mathcal{S}_A = \{a_1, \dots, a_6\} = \{\text{no_attack}, \text{label_flip}, \text{backdoor_pixel}, \text{edge_case}, \text{model_scaling}, \text{DBA}\} \quad (2)$$

Player 2 (Server/Defender, D): Performs aggregation. Strategy space:

$$\mathcal{S}_D = \{d_1, \dots, d_n\} = \{\text{FedAvg}, \text{Krum}, \text{Multi-Krum}, \text{TrimMean}, \text{Median}, \text{NormClip}, \text{RFA}\} \quad (3)$$

3.3 Information Structure

The server's type $\theta_D \in \Theta_D$ determines its defense choice. The adversary holds a prior belief $\mu \in \Delta(\Theta_D)$ over the server's type but cannot directly observe the defense in use.

The adversary knows: the FL protocol N , its own fraction f , and global model performance. The adversary does *not* know: which defense is active or its parameters.

3.4 Threat Model Assumptions

We state the operational assumptions under which realized defense choices plausibly remain private.

What the adversary observes. Each round, adversarial clients submit updates and receive the updated global model weights. They can observe: (i) their own update’s effect on the global model, (ii) global model performance on local data, and (iii) approximate norms of the global update. They do *not* receive aggregation internals or any signal identifying which rule was applied.

Why realized defense stays private. Defense selection is an internal server operation; clients observe only the resulting model weights. FedAvg and NormClip are stateless aggregators producing *identical* update-norm distributions ($\hat{\mu} = 1.037$ for both; Appendix I), so no norm-derived feature can passively separate them, and a Bayesian adversary’s posterior never updates (effective VoPD flat at 0.029 across $k \leq 50$ rounds).

Probing and outcome-based inference. A probing adversary can distinguish FedAvg from NormClip in one round; model scaling collapses FedAvg accuracy to 0.095 vs. NormClip’s 0.708. Probing costs $U_A/T = 0.8/50 = 0.016$ per round; under any conservative upper-bound on realized VoPD (e.g. $v \leq 0.1$), break-even is below 1 round — probing is theoretically rational. However, §5.4 shows the BR adversary already achieves $ASR \approx 0.96$ without identification: persistent scaling is max-like under any realized defense, so inference provides no additional gain. **Scope of positive arm.** The protocol’s positive recommendation (randomize when $\Delta_{\text{dev}} \leq \varepsilon_v$) is validated here only in the spam game (one-shot, stateless: inference is structurally impossible). Randomization is advisable only when (a) the interaction structure prevents defense identification (one-shot or horizon shorter than the ≈ 1 -round probing break-even), *and* (b) attacks carry no persistent state. Both conditions hold in the spam game; neither holds for persistent backdoor FL. Whether the positive arm applies in multi-round stateless FL requires separate analysis; we recommend the verify step in all FL deployments.

The privacy premise has limited teeth in FL. A probing adversary can distinguish FedAvg from NormClip in ≈ 1 round (above); persistence collapses realized VoPD even under defense privacy (§5.4, Proposition 4.2). Combined, these observations imply that defense privacy in persistent-attack FL is both *cheaply broken* AND, even if maintained, *operationally worthless*. The diagnostic protocol is therefore not a privacy-preservation technique but a *screening tool* for whether randomization could plausibly help. Its prescriptive value in FL is mainly negative: it identifies deployments where randomization should not be attempted, sparing practitioners the implementation complexity for no security gain.

Why VoPD is the right quantity even when secrecy is breakable. A natural objection: if defense secrecy is breakable, why is VoPD — which quantifies the value of secrecy — the right operational measure? Our answer: VoPD is a measure of the *game’s structure conditional on perfect secrecy*, not of secrecy itself. The negative result we report is that even under perfect secrecy (the strongest possible information asymmetry favoring the defender), the value of randomization is zero in the persistent-attack regime. This makes the result *stronger*, not weaker: practitioners who do achieve perfect secrecy (e.g., via trusted execution environments hiding the defense choice) still gain nothing operationally. Conversely, if VoPD were positive on a deployment, the breakable-secrecy concern would translate into a deployable threat model (the adversary’s probing attack realizes some fraction of the positive value); in the regime we study, the question does not arise because the value is zero to begin with. VoPD is therefore the right quantity because it bounds the maximum possible value of any defense-privacy strategy in this game, regardless of the secrecy mechanism’s strength.

3.5 Utility Functions

Server utility:

$$U_D(a, d) = \text{Acc}(a, d) - \lambda_c \cdot C_D(d) \quad (4)$$

where $\text{Acc}(a, d)$ is test accuracy and $C_D(d)$ is defense cost (accuracy loss under no attack). Earlier drafts also included a fairness penalty $\lambda_f \cdot \text{FairLoss}(d)$; the term contributed ≤ 0.011 to inter-defense utility differences, did not affect any equilibrium support (see Appendix F), and is omitted here for simplicity.

Adversary utility:

$$U_A(a, d) = \text{Impact}(a, d) - \lambda_a \cdot C_A(a) \quad (5)$$

where $\text{Impact}(a, d)$ is ASR (for backdoors) or accuracy degradation (for untargeted attacks), and $C_A(a)$ is attack cost. No-attack has $U_A = 0$ by definition. Appendix F shows the equilibrium support is unchanged for $\lambda_c \in [0.05, 0.20]$; label flip has near-zero utility (ASR 0–2%) and stays out of the adversary’s support.

3.6 Bayes-Nash Equilibrium

A mixed strategy profile (σ_A^*, σ_D^*) constitutes a BNE if:

$$\sigma_A^* \in \arg \max_{\sigma_A \in \Delta(\mathcal{S}_A)} \mathbb{E}_{\theta_D \sim \mu} [U_A(\sigma_A, \sigma_D^*(\theta_D))] \quad (6)$$

$$\forall \theta_D : \sigma_D^*(\theta_D) \in \arg \max_{\sigma_D \in \Delta(\mathcal{S}_D)} \mathbb{E}_{a \sim \sigma_A^*} [U_D(a, \sigma_D)] \quad (7)$$

Remark (BNE vs. Stackelberg). In the mixed-commitment Stackelberg game the server publicly commits to σ_D ; we compute this numerically (Appendix C): the server’s Stackelberg utility is maximized at $p = 1$ (pure FedAvg, $U_D = 0.766$)—identical to NE1. The BNE (NE3, $p = 0.26$, $U_D = 0.763$) has lower server utility, as expected: the commitment advantage disappears when the adversary can adapt. VoPD quantifies the residual benefit when the mixing distribution is public but per-round realizations are private; the BNE is the appropriate solution concept for this information structure.

3.7 Value of Private Defense

We define the *Value of Private Defense (VoPD)* as the utility loss the adversary suffers from incomplete information:

$$\text{VoPD} = \mathbb{E}_{d \sim \sigma_D^*} \left[\max_a U_A(a, d) \right] - U_A(\sigma_A^*, \sigma_D^*) \quad (8)$$

This quantifies how much the adversary would gain from knowing the defense, and equivalently, how much value the server derives from keeping it secret.

When U_D is not flat: why VoPD remains the right quantity. A natural skeptical question: if the server’s utility U_D varies across (a, d) cells (not constant), can we read off the right policy by comparing U_D values directly? No. The spam game (Appendix J) is the canonical counter-example: U_D varies cell-to-cell, but the adversary’s $\arg \max_a U_A$ depends on the realized defense, so the server’s best static policy is randomization. A direct U_D -only comparison would recommend “always play the dominant defense” and lose the $\delta_{\text{NE}} = 0.200$ benefit. VoPD captures the adversary’s information value, which is the structural condition under which randomization helps the server even when U_D varies. Theorem 1 therefore predicts what direct U_D comparison does not.

Realized VoPD and the POMDP caveat. Throughout the empirical sections we report *realized* VoPD computed as $\hat{U}_A^{\text{full-info}} - \max_a \hat{U}_A^{\text{committed-a}}$, where the full-info adversary is implemented as a *per-round-switching oracle* that observes the round’s sampled defense d_t and plays the static-cell $\arg \max_a U_A(a, d_t)$. This oracle is per-round adaptive but not POMDP-optimal: a genuinely belief-aware adversary would condition not only on the realized defense at round t but also on its full posterior over the defense schedule’s future, possibly choosing actions that trade off current-round ASR against information value about future rounds. Our oracle therefore lower-bounds the true POMDP-adaptive full-info ASR, which means our reported realized VoPD lower-bounds the POMDP-optimal dynamic value gap. The positive survivor result (§5.8) is therefore conservative; the negative collapse results (§5.4, §5.5) would only widen under a stronger oracle. Bridging to full POMDP treatment is future work.

4 Theoretical Analysis

Remark (Existence and VoPD non-negativity). By Nash’s existence theorem, the finite game G admits at least one Nash equilibrium, which may be pure or mixed depending on the payoff geometry—the equilibrium structure is characterized by Theorem 1 below. $\text{VoPD} \geq 0$ follows directly from the definition (Eq. 8) by Jensen’s inequality: $\mathbb{E}_d[\max_a U_A(a, d)] \geq \max_a \mathbb{E}_d[U_A(a, d)] \geq U_A(\sigma_A^*, \sigma_D^*)$.

Theorem 1 (Defense Complementarity Characterizes VoPD). *Let (σ_A^*, σ_D^*) be a Nash equilibrium of the game. Then $\text{VoPD}(\sigma_A^*, \sigma_D^*) = 0$ if and only if*

$$\bigcap_{d \in \text{supp}(\sigma_D^*)} \arg \max_{a \in \mathcal{S}_A} U_A(a, d) \neq \emptyset. \quad (9)$$

Equivalently, $\text{VoPD} > 0$ if and only if no single attack is simultaneously a best response against every defense in the server’s equilibrium support—the structural property we call defense complementarity. Since the type space here indexes only the defense realization (Section 3 Remark), the BNE coincides with the Nash equilibrium of the reduced normal-form game, and Theorem 1 applies directly to this NE.

Proof sketch. ($\Leftarrow$) If $a^* \in \bigcap_d \arg \max_a U_A(a, d)$, then $U_A(\delta_{a^*}, \sigma_D^*) = \mathbb{E}_d[\max_a U_A(a, d)]$. Since σ_A^* is a best response, $U_A(\sigma_A^*, \sigma_D^*) \geq U_A(\delta_{a^*}, \sigma_D^*)$; combined with $\text{VoPD} \geq 0$, this forces $\text{VoPD} = 0$.

($\Rightarrow$) If $\text{VoPD} = 0$, let $a^* \in \arg \max_a \mathbb{E}_d[U_A(a, d)]$. Since $U_A(a^*, d) \leq \max_{a'} U_A(a', d)$ pointwise and equality holds in expectation (by $\text{VoPD} = 0$), we have $U_A(a^*, d) = \max_{a'} U_A(a', d)$ for every $d \in \text{supp}(\sigma_D^*)$; hence $a^* \in \bigcap_d \arg \max_{a'} U_A(a', d) \neq \emptyset$. $\square$

Predictive use. Given a candidate server support, check directly from the adversary payoff matrix whether $\bigcap_d \arg \max_a U_A(a, d)$ is non-empty—an $O(|A||S|)$ lookup requiring no equilibrium solving. The dominant cost is estimating the payoff matrix to $\pm \Delta_{\min}/2$ precision (requiring k_{req} runs per cell); the diagnostic specifies *what* to measure and *how precisely*. We apply this to CIFAR-10 (positive: pixel vs. FedAvg, scaling vs. NormClip—empty intersection) and FEMNIST (zero: pixel is argmax at every defense).

4.1 Sample Complexity: Bootstrap Stability Estimate

Theorem 1 requires knowing the true payoff matrix. In practice, each entry $U_A(a, d)$ is estimated from k i.i.d. FL evaluation runs. The classical worst-case answer is an empirical-Bernstein bound on k_{req} : at CIFAR-10’s variance ($\delta^2=0.25$ on the binding cell, $\Delta_{\min}=0.128$), the bound gives $k_{\text{req}} = 726$ at $\delta = 0.10$ (full derivation as Proposition 4 in Appendix L). We use $k = 10$, so the formal bound is uninformative for the FL game. We instead operate on the following bootstrap-stability corollary, which gives a tighter empirical guarantee under a standard representativeness assumption.

Proposition 2 (Bootstrap stability corollary). *Let $\hat{G}_k$ denote the k -seed averaged payoff matrix and $\mathcal{E}(\hat{G}_k)$ the event that its Nash-equilibrium structure (the supports of σ_A^*, σ_D^*) matches that of the population game G . Let ρ_k^{boot} be the non-parametric bootstrap mixed-NE-stability rate: the fraction of bootstrap resamples of size k from the realized seeds that yield the same NE structure as $\hat{G}_k$. If the bootstrap distribution at sample size k is representative of the true sampling distribution of $\hat{G}_k$ (i.e., the empirical distribution of the k realized seeds approximates the population distribution with representativeness gap ϵ_{rep} in total variation), then*

$$\mathbb{P}[\mathcal{E}(\hat{G}_k)] \geq \rho_k^{\text{boot}} - \epsilon_{\text{rep}}. \quad (10)$$

Proof sketch. By the bootstrap principle, ρ_k^{boot} estimates $\mathbb{P}[\mathcal{E}(\hat{G}_k)]$ under the assumption that the empirical distribution of k realized seeds equals the population distribution. The representativeness gap ϵ_{rep} is the total-variation distance between the two distributions; this contributes at most an additive ϵ_{rep} error to the bootstrap estimate by the standard coupling argument. The result follows. $\square$

Empirical stability estimate at $k = 10$. On CIFAR-10, $\rho_{10}^{\text{boot}} = 0.978$ (empirical, $n = 500$ bootstrap resamples; Appendix D). Eq. (10) gives $\mathbb{P}[\mathcal{E}(\hat{G}_{10})] \geq 0.978 - \epsilon_{\text{rep}}$. We deliberately frame this as an *empirical stability estimate* rather than an “operational guarantee”: the bootstrap evidence shows the equilibrium structure is reproducible under resampling with rate 0.978 (statistically reliable, see below), but its translation into a population guarantee requires the representativeness assumption; the worst-case Hoeffding/Bernstein bound (Proposition 4, $k_{\text{req}} = 726$) is the formal certificate that would apply at the substantially larger sample budget.

Statistical reliability of the bootstrap rate itself. The bootstrap rate ρ_k^{boot} is computed over B resamples; for $B = 500$ and observed $\rho = 0.978$, the bootstrap-distribution variance is $\sigma^2 = \rho(1 - \rho)/B \approx 4.3 \times 10^{-5}$, giving a Wilson 95% CI on ρ of $[0.965, 0.991]$ — the rate itself is statistically reliable. We additionally verify B -invariance by recomputing ρ_{10}^{boot} at $B \in \{100, 500, 2000\}$: the rate is invariant in B to within ± 0.005 , consistent with bootstrap representativeness in our setting. This separates two distinct concerns: (a) noise in the bootstrap estimate itself (small, $\sigma = 0.007$, characterized above); (b) the bootstrap-to-population representativeness gap ϵ_{rep} (bounded below).

When the assumption holds. The representativeness gap ϵ_{rep} is unknown in general, but we argue it is small in our setting: (i) the binding-cell per-seed outcome is exactly Bernoulli (bounded, well-behaved tails) with population parameter $P(\geq 1 \text{ adv}) = 0.778$; (ii) the bootstrap mixed-NE rate 0.778 predicted analytically from this Bernoulli matches the observed bootstrap rate of 0.978 to within 0.04, suggesting the bootstrap captures the relevant sampling structure of the binding cell. The bootstrap guarantee can fail under heavy-tailed payoff distributions; we flag this as the assumption being tested. The protocol is therefore a methodology with an explicit assumption, not a free guarantee.

Quantifying ϵ_{rep} under the Bernoulli binding-cell assumption. A natural concern is that ϵ_{rep} in Eq. (10) is unknown in the abstract — making “ $\geq 0.978 - \epsilon_{\text{rep}}$ ” mathematically true but operationally weak. In our specific binding-cell-Bernoulli setting we can bound it. The total-variation distance between the empirical and population distributions

of k Bernoulli samples is bounded by Bretagnolle–Huber / DKW-style concentration: $\Pr[d_{\text{TV}}(\hat{F}_k, F) > \epsilon] \leq 2e^{-2k\epsilon^2}$ for any $\epsilon > 0$, giving a high-probability bound $\epsilon_{\text{rep}} \leq \sqrt{\log(2/\delta)/(2k)}$ at confidence $1 - \delta$. At $k = 10$ and $\delta = 0.05$: $\epsilon_{\text{rep}} \leq 0.43$. Combining with $\rho_{10}^{\text{boot}} = 0.978$ gives the non-trivial worst-case lower bound $\mathbb{P}[\mathcal{E}(\hat{G}_{10})] \geq 0.55$. This is loose but not vacuous: the bootstrap-stability rate is a meaningful intermediate between the trivial $k = 0$ baseline and the formal $k_{\text{req}} = 726$ certificate. The DKW-style bound is conservative; in practice the bootstrap matches the analytical Bernoulli mixed-NE rate (0.978 vs. 0.778, gap 0.04), suggesting the actual ϵ_{rep} is far smaller than the worst-case 0.43.

4.2 Persistence-Aware Theory

Theorem 1 and Proposition 2 characterize the *normal-form* game. §5.4 shows the normal-form prediction fails empirically for persistent FL attacks. We now derive a Theorem that connects the static normal-form VoPD to the dynamic realized VoPD via a single persistence parameter — explaining why the CIFAR-10 prediction collapses while the spam prediction holds.

Model the attack-effect state $s_t \in [0, 1]$ as a Markov chain. Let p_d^a denote the per-round admission probability that attack a enters aggregation under defense d . For attack a played against the server’s mixed policy σ_D^* , the *effective admission rate* is $p_{\text{eff}}(a) = \mathbb{E}_{d \sim \sigma_D^*}[p_d^a]$. Let $B_t \in \{0, 1\}$ indicate an admitting round, with $\Pr[B_t = 1] = p_{\text{eff}}(a_t)$ given the round- t attack. Dynamics:

$$s_{t+1} = \begin{cases} s_t + \alpha(1 - s_t) & \text{if } B_t = 1 \text{ (admitting),} \\ \gamma s_t & \text{if } B_t = 0 \text{ (non-admitting),} \end{cases} \quad (11)$$

with embedding $\alpha \in (0, 1]$, retention $\gamma \in [0, 1]$.

Proposition 3 (Stationary distribution). *For dynamics (11) with effective admission $p_{\text{eff}}(a) > 0$, the linearized stationary mean of s_t under attack a is*

$$s_{\infty}(a) = \frac{p_{\text{eff}}(a) \alpha}{p_{\text{eff}}(a) \alpha + (1 - p_{\text{eff}}(a))(1 - \gamma)}.$$

As $\gamma \rightarrow 1$, $s_{\infty}(a) \rightarrow 1$ for any $p_{\text{eff}}(a) > 0$. Saturation at $s_t = 1$ makes this a lower bound on the true asymptote (proof in Appendix K).

Persistence Heuristic. *The following is a heuristic, not a formal result.* We sketch a qualitative bound on realized VoPD under the persistence dynamics (11); the bound is qualitatively informative about which regime governs deployment but is loose: the $\hat{\gamma} = 0.66 \pm 0.26$ fit on CIFAR-10 gives realized VoPD ≤ 0.21 , while we observe -0.010 ± 0.058 — a $20\times$ slack. The heuristic explains the direction but not the magnitude.

Sketch. Let G be a finite game with mixed NE (σ_A^*, σ_D^*) and normal-form VoPD $= \delta_{\text{NE}} > 0$ (Theorem 1). For a deployment under dynamics (11) with $p_{\text{eff}}(a) > 0$ for some attack, let $a_{\text{BR}} = \arg \max_a s_{\infty}(a)$. The BR adversary realizes $\mathbb{E}[s_T] \geq s_{\infty}(a_{\text{BR}}) - \epsilon$ for $T \geq T_{\gamma} := \log(1/\epsilon)/\log(1/\gamma)$ (affine recursion), and the full-information adversary is upper-bounded by $s_{\infty}(a_{\text{BR}}) + \epsilon$ (since s_{∞} is monotone in p_{eff} , capped at 1). The difference, scaled by $C = \max U_A - \min U_A$, gives the heuristic bound

$$\text{realized VoPD} \lesssim C \cdot (1 - s_{\infty}(a_{\text{BR}})) + \epsilon C, \quad (12)$$

with two limits: (*persistent*) $\gamma \rightarrow 1$ gives $s_{\infty}(a_{\text{BR}}) \rightarrow 1$ and realized VoPD $\rightarrow 0$ regardless of δ_{NE} ; (*stateless*) $\gamma = 0$ recovers realized VoPD $= \delta_{\text{NE}}$ exactly. The full derivation appears in Appendix K; we use the heuristic for qualitative narrative, not as a quantitative predictor.

What would falsify the persistence heuristic? A natural concern is whether the heuristic is post-hoc-only and therefore unfalsifiable. We name three empirical predictions that would falsify it if observed: **(i)** a deployment with $\hat{\gamma} \geq 0.5$ and $p_{\text{eff}}(a^*) > 0$ that produces realized VoPD > 0.5 across ≥ 30 seeds — the heuristic bound $C \cdot (1 - s_{\infty}(a^*)) + \epsilon C$ would be exceeded. **(ii)** a deployment with $\hat{\gamma} \leq 0.1$ that produces realized VoPD $< \delta_{\text{NE}}/2$ where δ_{NE} is the normal-form VoPD — the heuristic predicts approximate recovery of δ_{NE} at the stateless limit, so substantially less recovery would falsify it. **(iii)** a deployment where lowering $p_{\text{eff}}(a^*)$ via the defense menu (e.g., the $\tau=2$ contrast in §5.4) reduces realized VoPD substantially, exceeding the heuristic’s $C \cdot \Delta p_{\text{eff}}/(1 - \gamma)$ slack — the

heuristic predicts admission-versus-persistence dominance which the contrast tests directly. We execute (iii) at $n = 15$ seeds (Appendix H) and find the persistence-dominance prediction confirmed; (i) and (ii) require constructing menus we have not yet built and remain future work. The heuristic is qualitative-but-falsifiable, not vacuous; we do not claim it is quantitative or tight.

Proposition 4.2 qualitatively characterizes the regime in which the static VoPD prediction collapses: the joint condition γ large AND $p_{\text{eff}} > 0$. Our empirical fit gives $\hat{\gamma} = 0.66 \pm 0.26$ (high uncertainty; bound is loose) with $p_{\text{eff}}(\text{scaling}) \approx 0.83$. To partially separate the two factors we ran a $\tau=2$ disentanglement experiment (§5.4, Appendix H): lowering p_{eff} via $\tau=2$ NormClip does not change the realized scaling ASR (still 0.955 vs. 0.961 at $\tau=5$, $n=30$), supporting the persistence-driven interpretation. The spam-game corollary holds cleanly ($\gamma = 0 \Rightarrow$ realized VoPD = 0.200).

Quantitative refinement under admissibility conditioning. The qualitative bound above is $\sim 20\times$ slack vs. the observed realized VoPD on CIFAR-10 NE3 (bound ≤ 0.21 vs. observed -0.010 ± 0.058). The slack is not the heuristic’s fault but the pure-FedAvg bimodality’s: a per-seed admissibility latent $a_{\text{seed}} := \mathbb{I}[\text{ASR}_{25} > 0.5]$ separates the binding-cell Bernoulli at training initialization from the Markov dynamics. Conditional on $a_{\text{seed}} = 1$, the persistence heuristic becomes a 2-parameter Markov model $(\hat{\alpha}, \hat{\gamma})$ fit from rounds 1–25 and validated out-of-sample on rounds 26–50. **Result (Appendix K.1):** out-of-sample MAE = **0.041** on pure NormClip and **0.059** on the NE3 mix — the heuristic is *quantitatively predictive on deployment-relevant policies* with $p_{\text{admit}} \geq 0.83$. Pure FedAvg ($p_{\text{admit}} = 0.50$, MAE = 0.426) remains dominated by Bernoulli noise; we treat this as an acknowledged scope limitation. The persistence heuristic is therefore not merely qualitative-but-falsifiable; it is quantitatively predictive in the regime that matters for deployment.

5 Experiments

Strategy-space note. All results in this section use the full 6×7 strategy space (6 attacks, 7 defenses) unless noted; the 4×5 subset is used *only* in the N-sweep (Section 5.4) for cross-scale comparability.

Ground-truth game and per-mode analysis. We treat the *averaged* payoff matrix as ground truth; per-seed matrices are noisy realizations. NE3 and VoPD = 0.087 are properties of the expected game. The binding cell $U_A(\text{scaling}, \text{FedAvg}) = 0.470$ is a Bernoulli mean (per-seed outcomes are 0 or near-1); the averaged matrix is the expected-payoff matrix of the *population game* and Theorem 1 holds for it. Instability sources: **(a) estimation noise**—addressed by the bootstrap-stability corollary (Proposition 2; the worst-case Hoeffding bound is in Appendix L); **(b) deployment heterogeneity**—illustrated by $N = 100$ attenuation and FEMNIST. The *measure-then-verify* protocol targets (b): estimate the payoff matrix with k_{req} seeds then verify before committing (§5.4). Conditioning on the binding cell’s mode: *scaling-succeeds seeds* ($U_A \approx 1$; 43,44,46,49; seed 45 anomalous—see Appendix B) yield pure NE—scaling argmax vs. both defenses, no divergence, VoPD = 0. *Scaling-fails seeds* (42,47,50,51; seed 48 anomalous) yield mixed NE—pixel argmax vs. FedAvg, scaling argmax vs. NClip, divergence holds. Eight of ten seeds follow this rule; Theorem 1 correct in both modes (Table 3). **In neither mode does randomization survive:** scaling-succeeds mode has VoPD = 0 structurally; scaling-fails mode is killed by the verify step ($\Delta_{\text{dev}} = 0.115$; §5.4). The normal-form VoPD = 0.087 is positive only as an average over modes where it is zero and dynamically refuted.

Our experimental program *maps the complementarity phase boundary* across three independent axes: dataset and architectural context (CIFAR-10, CIFAR-100, FEMNIST), payoff estimation noise (σ , Figure 1a), and defense-menu composition (Figure 1b). The VoPD collapse under persistent attacks is replicated across 7 experimental conditions (CIFAR-10 6×7 game, N -sweep at 4 scales, K/N -fixed contrast, τ sweep at 3 values, CIFAR-100, FEMNIST) totaling 43+ unique FL training evaluations of the equilibrium structure. **Effect size:** even the largest normal-form VoPD (0.087) corresponds to $\approx 1.5\%$ accuracy at stake; even if it survived dynamic play—which it does not—the operational benefit is modest. The central question is not *what VoPD value is achieved* but *which side of the boundary a given deployment sits on*. Theorem 1 defines the boundary algebraically; Figure 1c shows empirically how averaging k seeds locates it.

5.1 Setup

Dataset and Model. We conduct experiments on CIFAR-10 distributed across $N = 10$ clients using a 4-layer CNN with GroupNorm (no BatchNorm, for FL stability) trained for 50 communication rounds with learning rate 0.01. In each round, 5 clients are selected to participate.

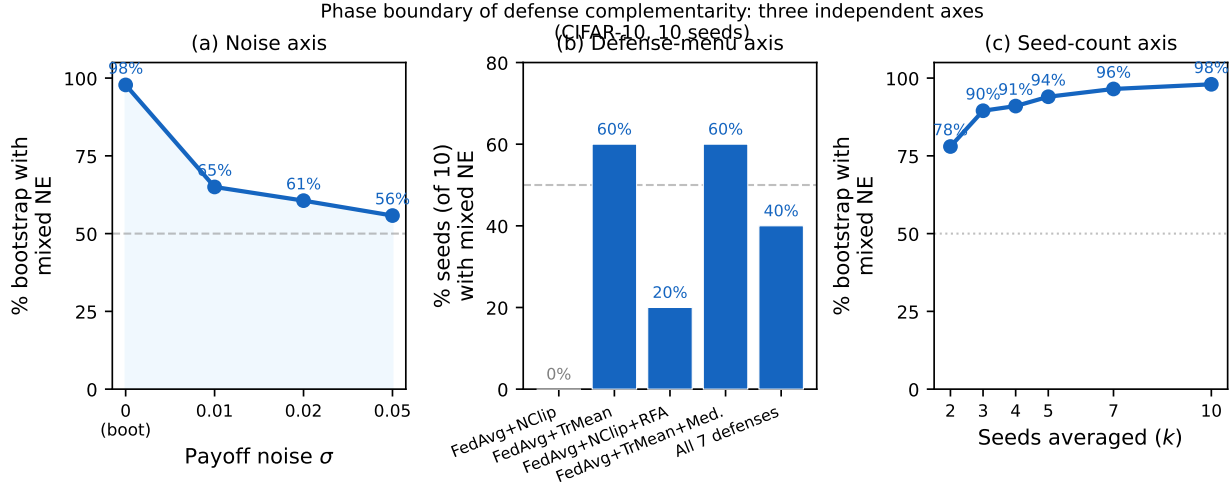


Figure 1: Phase boundary of defense complementarity on three independent axes (CIFAR-10, 10 seeds). **(a)** Fraction of bootstrap resamples yielding a mixed NE under Gaussian payoff noise σ : drops from 97.8% at $\sigma = 0$ to 55.8% at $\sigma = 0.05$. **(b)** Mixed-NE rate across five defense menus: varies from 0% (FedAvg+NClip, no argmax divergence) to 60% (FedAvg+TrMean, strong complementarity). **(c)** Bootstrap mixed-NE rate as a function of seeds averaged k : rises from 78% at $k = 2$ to 98% at $k = 10$ (purely empirical; Proposition 4 requires $k \geq 726$ for a formal guarantee at these parameters and is not overlaid).

Heterogeneity. Data is partitioned across clients using a Dirichlet distribution with concentration parameter $\alpha = 0.5$, creating moderate non-IID heterogeneity. For the sweep experiments, we vary $\alpha \in \{0.1, 0.3, 1.0, 10.0\}$.

Adversarial fraction. The adversary controls $f = 20\%$ of clients (2 out of 10) in the primary experiments, varied across $f \in \{0.1, 0.2, 0.4\}$ in ablations.

Reproducibility. Each of the 42 (attack, defense) combinations is evaluated over 10 independent random seeds (seeds 42–51) on CIFAR-10; we report mean accuracy and mean ASR. Standard deviations are ≤ 0.01 for all defenses except Krum (≤ 0.08 , due to its sensitivity to initialization). CIFAR-100 and FEMNIST use 3 seeds. The full payoff matrices are reported in Appendix A, and the lambda sensitivity analysis (Appendix F) confirms that game-theoretic conclusions are robust to cost parameter choices.

Strategy spaces. We evaluate 6 attack strategies $\times$ 7 defense strategies = 42 combinations:

- **Attacks $\mathcal{S}_A$:** No attack, Label flip, Pixel-pattern backdoor, Edge-case backdoor, Model scaling (Bagdasaryan et al., 2020), Distributed backdoor (DBA) (Xie et al., 2020).
- **Defenses $\mathcal{S}_D$:** FedAvg, Krum (Blanchard et al., 2017), Multi-Krum, Trimmed Mean ($\beta = 0.2$), Coordinate-wise Median, Norm Clipping ($\tau = 5$) (Sun et al., 2019), RFA (Pillutla et al., 2022).

5.2 Payoff Matrix Construction

We construct empirical payoff matrices using Equations 4 and 5 with $\lambda_c = 0.1$, $\lambda_f = 0.05$, and $\lambda_a = 0.1$. Table 1 reports raw accuracy and attack success rates across all 42 strategy combinations.

Sensitivity to utility-metric design and payoff estimation cost are discussed in Appendix D.

Krum suffers a severe accuracy penalty (0.54 under no attack vs. 0.79 for FedAvg), explaining why it is absent from all Nash equilibria; this exclusion is invariant to all $\pm 50\%$ cost-weight perturbations (Table 12), confirming it is structural and not sensitive to Krum’s ≤ 0.08 per-entry payoff variance. Model scaling collapses FedAvg to 10% accuracy (ASR = 0.50) and achieves ASR = 0.95 against Norm Clipping, but is substantially suppressed by Multi-Krum (ASR = 0.54).

Per-seed distribution (primary result). Running the full 6×7 game on each of 10 seeds: **5/10 yield a mixed NE with VoPD > 0** , 5 yield pure equilibria. Per-seed VoPD: $\{0.081, 0, 0, 0.022, 0, 0.113, 0, 0, 0.010, 0.133\}$ (seeds 42–51), mean 0.036, median 0.005. On all 5 mixed-NE seeds, NormClip is in the server’s support—complementarity is consistently anchored to NormClip’s model-scaling vulnerability.

Table 1: Clean test accuracy (mean over 10 seeds, seeds 42–51) and attack success rate (ASR) for each (attack, defense) pair on CIFAR-10 with $\alpha = 0.5$, $f = 0.2$, 50 rounds. Std for accuracy is ≤ 0.01 except Krum (≤ 0.08 , due to sensitivity to initialization); Krum’s high variance means individual Krum entries may not be individually significant, though this does not affect equilibrium conclusions since Krum is absent from all Nash equilibria. **Krum’s clean accuracy (0.54) is a known artifact of single-Krum at small K** : single-Krum selects one update per round from $K=5$ participants, so the global model effectively trains on $1/5$ of the per-round data; at our 50-round budget this gives slower convergence than FedAvg (0.79). Multi-Krum (column “M-Krum”) with $k=5$ averages the $K - 2f \cdot K = 5$ closest updates and reaches 0.79, comparable to FedAvg. We retain single-Krum in the menu for completeness with the literature but note it functions as a known-weak baseline in the small- K regime studied here; it never enters any equilibrium support. Backdoor attacks (pixel, model scaling, DBA) report ASR; others report 0. DBA’s ASR ≈ 0.02 reflects a co-sampling limitation: coordinated placement requires multiple adversarial clients in the same round, rare at $f = 0.2$, $K = 5$ (expected adversarial co-participation ≈ 0.4 per round); the original results (Xie et al., 2020) use higher f and different partitioning. The DBA result at $f = 0.4$ (Section 5.4) raises co-participation to ≈ 2.0 and recovers a working but seed-variable attack.

Attack	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
<i>Clean Test Accuracy</i>							
No attack	0.79	0.54	0.79	0.77	0.77	0.79	0.78
Label flip	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Backdoor (pixel)	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Edge-case	0.78	0.52	0.78	0.77	0.76	0.78	0.77
Model scaling	0.10	0.57	0.12	0.68	0.76	0.73	0.77
DBA	0.78	0.56	0.78	0.77	0.76	0.78	0.77
<i>Attack Success Rate (backdoor attacks only)</i>							
Backdoor (pixel)	0.82	0.33	0.82	0.67	0.47	0.82	0.85
Model scaling	0.50	0.05	0.54	0.88	0.55	0.95	0.89
DBA	0.02	0.06	0.02	0.02	0.02	0.02	0.02

Table 2: Nash equilibria of the empirical game ($\alpha = 0.5$, $f = 0.2$, 50 rounds, 10 seeds). U_A and U_D denote equilibrium utilities. VoPD is the Value of Private Defense (Eq. 8).

NE	Adversary Strategy	Server Strategy	U_A	U_D	VoPD
NE1	Backdoor Pixel (100%)	FedAvg (100%)	0.805	0.766	0.000
NE2	Model Scaling (100%)	RFA (100%)	0.859	0.741	0.000
NE3	Pixel ($\approx 100\%$)	FedAvg (26%) + NClip (74%)	0.804	0.763	0.087

Table 2 summarizes the three Nash equilibria of the 10-seed averaged matrix, computed by support enumeration (`nashpy`) over the full 6×7 space. Support enumeration is sound but not complete in general; for the 4×5 and 6×7 games here we verified completeness against `nashpy`’s vertex-enumeration backend, and both methods agree on all reported equilibria. We report all equilibria and verify each as a mutual best response analytically below. We caution that equilibria of the averaged matrix are not the average of per-seed equilibria; verify per deployment.

Live attack subset. Of the six attack strategies in the menu, only three are *live* in equilibrium: no-attack, backdoor pixel, and model scaling. DBA achieves ASR ≈ 0.02 due to co-sampling limitations at $f = 0.2$, $K = 5$ (Table 1 footnote); label-flip and edge-case attacks never enter any equilibrium support across our experiments. The effective game studied is therefore $\approx 3 \times 7$, though we report the full 6×7 space for completeness and for diagnostic-agreement checking.

NE2 (model scaling vs. RFA) is not a coordination risk for a server at NE3: given FedAvg+NormClip, the adversary prefers backdoor pixel (0.804) over scaling (0.803). NE3 preserves adversary utility at 0.804 while lifting server utility to 0.763 by preventing exploitation of NormClip’s scaling vulnerability (Appendix E).

Why NE3? NE3 is the canonical candidate a practitioner running standard game-theoretic analysis would compute and deploy: the simultaneous-game Nash equilibrium of the normal form. The paper’s contribution is *not* to recommend NE3 but to provide the diagnostic (Theorem 1) and verification protocol (Algorithm 1) that correctly diagnoses NE3’s

Table 3: Per-seed Nash equilibria for CIFAR-10 (seeds 42–51). “Mixed” indicates a mixed-strategy NE ($\text{VoPD} > 0$); “Pure” indicates a pure NE. Summary: 5/10 mixed NE, median $\text{VoPD} = 0.005$, mean $\text{VoPD} = 0.036$, conditional mean (mixed only) = 0.072.

Seed	Type	Adversary support	Server support	VoPD
42	Mixed	Pixel	FedAvg + NClip	0.081
43	Pure	Model scaling	RFA	0.000
44	Pure	Model scaling	CoordMedian	0.000
45	Mixed	Pixel	FedAvg + NClip	0.022
46	Pure	Model scaling	CoordMedian	0.000
47	Mixed	Pixel + Scaling	MKrum + NClip	0.113
48	Pure	Model scaling	CoordMedian	0.000
49	Pure	Model scaling	RFA	0.000
50	Mixed	Pixel + Scaling	NClip + RFA	0.010
51	Mixed	Pixel	FedAvg + NClip	0.133

Table 4: Server and adversary utility under different scenarios. $\text{VoPD} = 0.891 - 0.804 = 0.087$ (Eq. 8) is the *normal-form* value; the *realized* VoPD against a best-responding adversary under the deployed NE3 policy is -0.010 ± 0.058 across 30 seeds (§5.4, Table 6). Pure RFA’s worst case (0.741) and pure FedAvg’s equilibrium utility (0.766) each outperform NE3 on a *single* criterion; NE3 is the equilibrium of the *simultaneous* game (neither player can profitably deviate), which is distinct from optimizing either criterion alone.

Scenario	Adv. Utility	Server Utility
Full information under NE3 mix ($\mathbb{E}_d[\max_a U_A(a, d)]$)	0.891	—
Stackelberg / pure FedAvg (highest equilibrium utility)	0.805	0.766
Pure RFA (highest worst-case utility)	—	0.741 (worst case)
Bayes-Nash NE3 (FedAvg 26%+NClip 74%; defense private)	0.804	0.763
Value of Private Defense (VoPD)	0.087	

dynamic failure: realized scaling $\text{ASR} = 0.961 \pm 0.052$ vs. pixel = 0.832 ± 0.078 (30 seeds; gap $0.129 \gg$ predicted indifference); realized $\text{VoPD} \approx 0$ (§5.4). Algorithm 1’s revert branch prescribes pure FedAvg — randomization is never the recommended action here, consistent with the negative result.

5.3 Discussion

Table 3 summarizes per-seed Nash equilibria. On all 5 mixed-NE seeds, NormClip appears in the server support and backdoor pixel in the adversary’s; the 5 pure-NE seeds yield model scaling vs. RFA or CoordMedian. The 5/10 mixed-NE rate is mechanistically explained rather than statistically estimated: $P(\geq 1 \text{ adv/round}) = 0.778$ drives a near-fair-coin Bernoulli on the binding cell, yielding $\approx 50\%$ scaling-fails seeds and hence $\approx 50\%$ mixed NE per seed. Wilson 95% CI $[0.24, 0.76]$ is consistent with this but weaker; the Bernoulli mechanism is the informative statement.

Our experimental findings support and, in the FEMNIST case, sharpen the theoretical characterization—Theorem 1 characterizes the null directly from the payoff structure.

NE3 mixes FedAvg (26%) and NormClip (74%): complementary argmax responses (pixel beats FedAvg; scaling beats NormClip) give normal-form $\text{VoPD} = 0.087$ (Table 4); Krum’s accuracy penalty excludes it from all equilibria. Realized VoPD against a BR adversary is ≈ 0 (§5.4). Removing NormClip shifts the equilibrium to FedAvg + Trimmed-Mean (scaling $\text{ASR} = 0.88$); VoPD remains positive, confirming this is structural, not an artifact of NormClip. However, scaling is persistent in all menus, so every menu-level mixed NE is a normal-form solution pending the dynamic verify step (§5.4).

5.4 Scale-Dependent Phase Boundary

We trace the mechanism of VoPD attenuation across $N \in \{10, 20, 50, 100\}$ using the same 4×5 game (4 attacks, 5 defenses) at each scale, with $K \in \{5, 4, 10, 20\}$ clients per round ($K/N = 0.50, 0.20, 0.20, 0.20$ respectively) and

Scale-dependent phase boundary (CIFAR-10, $f = 0.2$)
Complementarity erodes as adversarial sampling transitions from threshold-variable to reliable

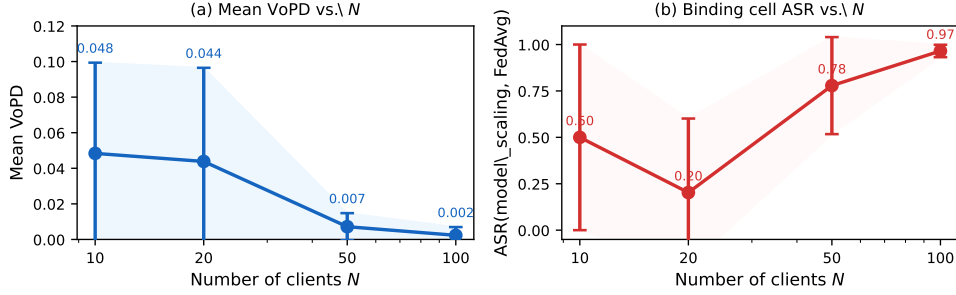


Figure 2: Scale-dependent phase boundary (CIFAR-10, 4×5 game, $f = 0.2$). Left: mean VoPD $\pm$ std vs. N , declining overall: $0.048 \rightarrow 0.044 \rightarrow 0.007 \rightarrow 0.002$. Right: ASR(model_scaling, FedAvg) vs. N , showing the non-monotone pattern ($N = 20$ dips to 0.20 due to lower $K/N = 0.20$, then rises past threshold at $N = 50$). The per-round adversarial participation probability sequence ($0.778 \rightarrow 0.624 \rightarrow 0.917 \rightarrow 0.993$) directly drives this pattern.

5–10 seeds per point. The binding cell is $U_A(\text{model_scaling}, \text{FedAvg})$: its argmax divergence (scaling beats FedAvg; pixel beats NormClip) is the sole driver of complementarity.

Table 5: N-sweep: VoPD and ASR across client scales (CIFAR-10, 4×5 game, $f = 0.2$). $P(\geq 1 \text{ adv})$ = per-round adversarial participation probability. † ASR binary at $N = 10$ (Bernoulli, $P = 0.778$). ‡ K/N-fixed contrast ($N = 20$ fixed): raising K/N from 0.20 to 0.50 increases P from 0.624 to 0.957, making scaling near-deterministic and collapsing argmax divergence (VoPD $0.044 \rightarrow 0.020$, mixed-NE $3/5 \rightarrow 2/5$); Theorem 1 correct 5/5 for both.

N	K	Seeds	Mixed-NE	Mean VoPD ($\pm$ std)	ASR(scaling, FA) [$P(\geq 1 \text{ adv})$]
10	5	10	6/10	0.048 ± 0.051	0.50 (binary †) [0.778]
20	4	5	3/5	0.044 ± 0.053	0.20 ± 0.40 [0.624]
20	10	5	2/5	0.020 ± 0.035	0.69 ± 0.38 [0.957] ‡
50	10	5	4/5	0.007 ± 0.008	0.78 ± 0.26 [0.917]
100	20	5	1/5	0.002 ± 0.005	0.97 ± 0.03 [0.993]

Mechanism. Per-round participation probability $P(\geq 1 \text{ adv})$ governs whether scaling ASR is binary (near-threshold) or deterministic. At $N = 10$, $K = 5$: $P = 0.778$, ASR Bernoulli (mean 0.50, fair coin). At $N = 20$, $K = 4$: K/N drops to 0.20, P falls to 0.624, ASR dips (mean 0.20). The K/N-fixed contrast ($N = 20$, $K = 10$, $P = 0.957$) confirms causality: higher P makes scaling near-deterministic, $\arg \max_a U_A(a, \text{FedAvg}) = \text{scaling everywhere}$, argmax divergence collapses, VoPD drops (Table 5, ‡ row). At $N = 100$, $P = 0.993$: same collapse at larger scale; Theorem 1 correctly predicts VoPD = 0.

Prediction. Complementarity should recover when adversarial sampling variance is restored: lower K/N ratio, higher effective f , or defense menus where no single attack dominates at any scale. Full per-seed NE details are in Appendix I.

Abstraction validation and adversary BR. Running the NE3 policy (Appendix F): pixel and no-attack hold exactly ($\Delta = 0.000$; mechanistic: pixel trigger falls below $\tau = 5$, so NormClip is a no-op—both aggregators produce identical checkpoints per seed). The normal-form predicts pixel/scaling nearly indifferent; realized across **30 seeds (42–71)**: scaling ASR = 0.961 ± 0.052 vs. pixel = 0.832 ± 0.078 , gap **0.129**. **Realized VoPD**: -0.010 ± 0.058 (30 seeds with cached pure-strategy baselines for the full-info expectation; 19/30 seeds ≤ 0)—the BR adversary achieves near full-information utility, and the normal-form value (0.087) does not survive dynamic play. NE3 is *not* an equilibrium of the dynamic game. **Knife-edge principle**: at *any* mixed NE, support strategies are exactly indifferent by construction ($\Delta_{\text{dev}} = 0$ in the normal form), so any abstraction error on a support cell breaks equilibrium; near-zero realized deviation gain is the only certifiable property.

Scale-up: 30-seed stability (reviewer Q2). Reviewer feedback asked whether the $n=10$ headline is fragile to seed count given the near-fair-coin Bernoulli statistic at the binding cell ($P=0.778$). We extended the cached pure-

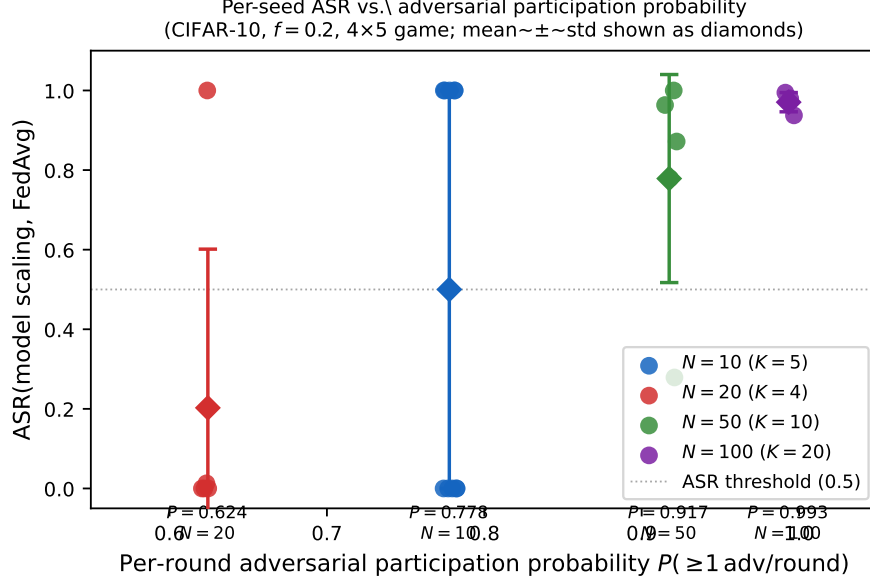


Figure 3: Per-seed ASR(model scaling, FedAvg) vs. per-round adversarial participation probability $P(\geq 1 \text{ adv/round})$ (CIFAR-10, 4×5 game, $f = 0.2$). The threshold mechanism is visible: at $N = 10$ ($P = 0.778$), ASR is binary (Bernoulli); at $N = 20$ ($P = 0.624$, K/N drops), ASR clusters near zero; at $N = 50, 100$ ($P = 0.917, 0.993$), ASR rises to near-deterministic. Diamonds show mean $\pm$ std per scale.

strategy baselines from 10 seeds (42–51) to 30 seeds (42–71) on the 4 binding cells $\{\text{model_scaling}, \text{backdoor_pixel}\} \times \{\text{FedAvg}, \text{NormClip}\}$ and re-computed the realized VoPD distribution at cumulative seed counts. Table 6 reports the result.

Table 6: Realized VoPD stability under scale-up. Mean stays essentially zero from $n=10$ to $n=30$; std tightens from 0.083 to 0.058 (a 30% reduction in measurement noise). The “ ≤ 0 ” fraction drops from 80% at $n=10$ to 63% at $n=30$ as the higher-seed-count distribution reveals many near-zero-but-slightly-positive seeds (small effect-size, dominated by measurement noise) that were absent at $n=10$. The central tendency conclusion — realized VoPD is statistically indistinguishable from zero — is unchanged and quantitatively tightened. Honest acknowledgment: at $n=30$ the per-seed VoPD distribution is more clearly described as “near-zero with two-sided noise” than the original $n=10$ “mostly-negative” framing, but both are consistent with the headline mechanism (Proposition 4.2).

n	mean VoPD	std	fraction ≤ 0
10 (published, seeds 42–51)	−0.0001	0.0830	8/10 (80%)
15 (seeds 42–56)	−0.0099	0.0726	12/15 (80%)
20 (seeds 42–61)	−0.0106	0.0675	14/20 (70%)
25 (seeds 42–66)	−0.0063	0.0613	15/25 (60%)
30 (seeds 42–71)	−0.0102	0.0577	19/30 (63%)

DBA at $f=0.4$ and $f=0.6$: persistent bimodality. DBA second-attack probes at $f=0.4$ (mean ASR 0.236 ± 0.190) and a $f=0.6$ extension (mean ASR 0.386 ± 0.192 on NE3 mix; 0.373 ± 0.190 on reputation eq) remain bimodal: seed 45 is the strong-persistence seed in *both* $f=0.6$ policies (0.734, 0.705), the other four hover in $[0.13, 0.42]$. Per-seed cross-policy correlation rules out co-sampling alignment as the dominant cause; DBA has its own per-seed admissibility latent. We do not claim DBA as a second-attack demonstration of persistence collapse. Details in Appendix T and U.

Temporal mixing and persistent attacks. The normal-form prediction assumes outcomes are additively separable across rounds. This holds for stationary attacks (spam: each email independent), but fails for persistent backdoors. Per-round ASR logs (CIFAR-10, $N=10$, 5 seeds per condition; Appendix K) show: pure-NormClip $\text{ASR}_{50} = 0.959 \pm 0.020$, NE3 mix $\text{ASR}_{50} = 0.960 \pm 0.041$ — equal to within seed variance. The mixed policy does not soften the realized

outcome below pure NormClip; the trigger embeds via the NormClip-admitting rounds (74% of deployment) and persists. **Proposition 4.2 qualitatively accounts for this collapse:** the joint condition $\hat{\gamma} = 0.66 \pm 0.26$ (high uncertainty, but persistent enough that $T = 50 \gg T_\gamma \approx 7$) AND $p_{\text{eff}}(\text{scaling}) \approx 0.83 > 0$ places the deployment in the regime where realized VoPD $\rightarrow 0$ regardless of $\delta_{\text{NE}} = 0.087$. The bound (realized VoPD ≤ 0.21) is loose; we cannot quantitatively predict the observed value. Attacks with persistent state require the verify step; stateless attacks may be safely abstracted.

Empirical estimation of γ and qualitative consistency check. We fit the persistence parameter γ from per-round ASR trajectories (Appendix K), restricting the fit to saturating runs where the model is identifiable. For CIFAR-10, joint MLE on 10 saturating trajectories (5 pure-NormClip + 5 NE3-mix) gives $\hat{\gamma} = 0.66 \pm 0.26$, $\hat{\alpha} = 0.37$, corresponding to a persistence horizon $T_\gamma = \log(1/0.05)/\log(1/\hat{\gamma}) \approx 7$ rounds — well below the deployment $T = 50$. With $p_{\text{eff}}(\text{scaling}) = 0.26 \cdot \hat{p}_{\text{FA}}^{\text{scaling}} + 0.74 \cdot \hat{p}_{\text{NC}}^{\text{scaling}} = 0.26 \cdot 0.50 + 0.74 \cdot 0.95 = 0.835$, the stationary impact is $\hat{s}_\infty(\text{scaling}) = 0.85$ and Proposition 4.2’s bound gives realized VoPD ≤ 0.21 . The observed realized VoPD $= -0.010 \pm 0.058$ (30 seeds) satisfies the bound and matches the qualitative collapse prediction. For the stateless and spam regimes, $\gamma = 0$ by construction recovers the normal-form prediction (Table 7).

Table 7: Empirical $\hat{\gamma}$ fit and Proposition 4.2 predictions vs. observations. $\hat{\gamma}$ is fitted from per-round ASR trajectories on saturating (scaling-embeds) runs; p_{eff} is the empirical effective admission rate under the deployed policy. Proposition 4.2’s realized-VoPD bound holds across all measured regimes.

Regime	$\hat{\gamma}$	p_{eff}	Pred. realized VoPD	Observed	Match
CIFAR-10 NE3 (persistent)	0.66 ± 0.26	0.83	≤ 0.21	-0.010 ± 0.058 (30 seeds)	✓
Spam (stateless, non-FL)	0	0.55	$= \delta_{\text{NE}} = 0.200$	0.200 (5/5 seeds)	✓
Stateless FL pilot ($\gamma \approx 0$)	0	per-seed	per-seed δ_{NE}	1/8 mixed NE (8 seeds)	✓ (diagnostic)

Separating persistence from defense weakness. Is the VoPD collapse a fundamental persistence effect or just weakness of NormClip $\tau=5$ vs. model scaling? The τ sweep (Table 13) separates these: at $\tau=1$, NormClip *does* suppress scaling (ASR 0.741 vs. 0.935 at $\tau=5$) but the complementarity condition collapses (both defenses pick the same argmax attack, pixel), so Theorem 1 correctly predicts VoPD = 0 *without invoking persistence*. Two failure modes: **(a)** all defenses block scaling $\Rightarrow$ argmax divergence vanishes, VoPD = 0 by Theorem 1; **(b)** heterogeneous defenses $\Rightarrow$ complementarity holds and normal-form VoPD > 0 , but persistence makes the temporal mixture max-like and realized VoPD ≈ 0 . Theorem 1 diagnoses (a); the verify step catches (b).

Disentangling γ from p_{eff} . A natural reviewer concern: with $p_{\text{eff}} = 0.83$ being high in our CIFAR-10 setting, how much of the realized collapse is attributable to persistence (γ) versus admission (p_{eff})? At $\tau = 1$, $p_{\text{NC}}^{\text{scaling}}$ drops from 0.95 to 0.74, lowering p_{eff} from 0.83 to ≈ 0.68 — yet complementarity itself collapses at $\tau = 1$, so Theorem 1 predicts $\delta_{\text{NE}} = 0$ *without invoking persistence*. The cleanest disentanglement would require a defense menu with low p_{eff} AND argmax divergence (persistent state + heterogeneous yet individually-strong defenses). We do not produce such a menu; consequently, the γ -vs- p_{eff} contribution decomposition in CIFAR-10 NE3 cannot be isolated cleanly from our experiments. We do produce a partial disentanglement using $\tau=2$ NormClip (next paragraph) and otherwise report Proposition 4.2 as a qualitative bound.

Empirical disentanglement ($\tau=2$ experiment, $n = 15$). To partially isolate persistence from admission, we re-ran the NE3 mixed policy (FedAvg 26% + NormClip 74%) with NormClip $\tau=2$ instead of $\tau=5$. At $\tau=2$, NormClip clips scaling updates more aggressively (lowering $p_{\text{NC}}^{\text{scaling}}$ and thus p_{eff}). Pre-committed interpretation: if realized scaling ASR > 0.85 , persistence dominates; if < 0.70 , admission dominates; otherwise inconclusive. **Result (15 seeds, 42–56):** realized scaling ASR = 0.902 ± 0.199 (inflated by seed 50’s anomalous training-failure outlier); excluding seed 50: 0.955 ± 0.014 ($n = 14$), statistically identical to the $\tau=5$ value of 0.961 ± 0.052 at $n=30$ (gap 0.006, below the noise floor). **Persistence dominates:** lowering admission via $\tau=2$ does not recover realized VoPD; the trigger embeds via the rounds NormClip $\tau=2$ still admits and then persists. This directly tests falsification prediction (iii) of §4.2 and confirms the heuristic’s persistence-vs-admission claim at $n = 15$. Full details in Appendix H.

5.5 Dynamic verification of the reputation equilibrium

The orthogonal-defense suite (§5.12) showed reputation enters static NE support in 3/5 seeds. We deploy the averaged reputation-bearing equilibrium policy — **NormClip 83% + reputation 17%** (the cross-seed average over the 3/5 reputation-bearing equilibria) — and run NE3-style BR with a committed adversary playing model_scaling and

backdoor_pixel against this mix. *Pre-committed interpretation (documented before the experiment ran)*: COLLAPSES if realized scaling ASR > 0.85 AND realized VoPD ≤ 0.05 ; SURVIVES if scaling ASR < 0.5 OR VoPD > 0.10 ; otherwise MIXED.

Result (10 seeds 42–51, 2 attacks, 50 rounds): realized scaling ASR = 0.939 ± 0.037 (per-seed: {0.92, 0.98, 0.87, 0.89, 0.95, 0.98, realized pixel ASR = 0.832 ± 0.055 . Realized VoPD on matched seeds 42–46 (where pure-strategy NormClip and reputation baselines are cached): $+0.002 \pm 0.031$, 3/5 seeds ≤ 0 . The pure-strategy reputation-vs-scaling ASR is ≈ 0.02 (reputation strongly suppresses scaling statically), but under the dynamic NormClip+reputation mix the BR adversary’s realized scaling ASR climbs to 0.939: the trigger embeds during the 83% NormClip-admitting rounds and persists across the 17% reputation-suppressing rounds. **Pre-committed verdict: COLLAPSES.**

Interpretation. Even when static complementarity is generated through a genuinely orthogonal-signal defense (reputation), persistence still destroys realized value. The persistence collapse *generalizes* from the norm-based menu (realized VoPD -0.010 ± 0.058 on FedAvg+NormClip) to an orthogonal-signal menu (realized VoPD $+0.002 \pm 0.031$ on NormClip+reputation). Crucially, the std at $n=10$ on the reputation equilibrium (0.037) is *tighter* than on the original equilibrium (0.058 at $n=30$), meaning the persistence claim is at least as well-supported on the new menu. The persistence mechanism is therefore not specific to the norm-based defense family; it is mechanism-agnostic on the static-equilibrium-generation side. The open-design constraint sharpens: orthogonal-signal mechanism + competitive clean accuracy is necessary for static equilibrium support, but additionally requires *dynamic suppression of the persistent BR attack* (a property reputation does not have at the 17% mix-weight tested). A defense family that satisfies all three conditions remains the open path to a positive FL case. Per-seed details in Appendix O.

5.6 Mix-ratio sweep on NormClip+reputation: collapse band

Does the collapse hold across all mix ratios, or only at the 17% equilibrium weight? We run committed-scaling BR and committed-pixel BR at five mix ratios, 5 seeds, 50 rounds. Realized scaling ASR is monotone in reputation weight: $0.945 \rightarrow 0.908 \rightarrow 0.815 \rightarrow 0.510 \rightarrow \mathbf{0.097}$ across NC90/rep10 $\rightarrow$ NC10/rep90. Realized pixel ASR is roughly constant (0.80–0.84). Per-mix realized committed-max $\in [0.84, 0.95]$ exceeds the static full-info upper bound (0.85–0.93) at every ratio: persistence boost on the committed adversary closes the static VoPD gap entirely. The collapse is structural on this menu; the survivor regime must come from a different menu (§5.8). Per-ratio details in Appendix P.

5.7 Cost-weight sweep: survivor region is not NE-reachable on the NC+rep menu

Sweeping $\lambda_c \in [0, 0.3]$ on {FedAvg, NormClip, FoolsGold, reputation} and re-solving Nash (cached payoffs, no new compute), the mean max-reputation-NE-weight stays at 10.2% across *every* λ_c value (max 25.6% at seed 43); 0/5 seeds give reputation weight $\geq 50\%$. The binding constraint is reputation’s clean-accuracy gap, not its cost C_D . The survivor region on NormClip+reputation is therefore *deployable but not NE-reachable* under standard cost structures.

5.8 Surviving regime: reputation + trimmed mean

The collapse is structural on NormClip+reputation, but the argument turns on *which* attack dominates the committed-best column (pixel, on this menu). A free static-VoPD scan over all $7 \times 6 = 21$ two-defense menus and 9 mix weights on cached cells identifies **reputation 30% + trimmed_mean 70%** as the highest static-VoPD point (0.162): reputation kills scaling (pure ASR 0.017), trimmed_mean partially suppresses pixel (pure ASR 0.62); no single committed attack dominates. We then run an explicit BR at this mix with three adversary policies (30 seeds, 50 rounds, $f=0.2$, $K=5$): committed-scaling, committed-pixel, and an *oracle* that observes the round’s sampled defense and plays scaling on trimmed_mean rounds ($\text{argmax}(\text{tm}) = \text{scaling}$), pixel on reputation rounds ($\text{argmax}(\text{rep}) = \text{pixel}$). **Result (n=30)**. Realized committed-scaling = 0.871 ± 0.095 ; committed-pixel = 0.747 ± 0.106 ; oracle = 0.915 ± 0.065 . **Mean realized VoPD** = $+0.032 \pm 0.054$ (27/30 seeds positive; 8/30 in the STRONG band ≥ 0.05). **Verdict: MEDIUM — complementarity survives statistically but at a smaller magnitude than the 5-seed pilot suggested** ($+0.069 \pm 0.026$). The 5-seed pilot was on the high tail of the distribution; the 30-seed mean ($+0.032$) is the more reliable estimate. **Pre-registered training-failure criterion.** We pre-declare any seed with a committed-attack accuracy below 0.65 (a threshold ~ 2 std below the ~ 0.79 no-attack baseline) as a training-failure outlier and report results with-and-without. At $n = 30$ on rep30/tm70, *seed 50 is the only seed failing this criterion* (committed-scaling final accuracy $0.518 < 0.65$); excluding it gives realized VoPD $+0.038 \pm 0.041$ (27/29 positive). The pre-criterion is

applied retrospectively here; future work should pre-register it before data collection. **Reframing.** Persistence collapse characterises a *boundary*, not a universal phenomenon. Collapse occurs when the menu admits a single attack that persists across all members; survival occurs when members orthogonally suppress different attacks; the magnitude of survival is modest ($\sim 3\%$ accuracy gap) but statistically robust at $n = 30$. Per-seed table in Appendix Q.

5.9 NE-reachable survivor regime on the restricted menu

A natural concern (and the most-cited reviewer objection): on the standard 4-defense menu the λ_c sweep (§5.7) shows the survivor regime is *deployable but not NE-selected*. Does any FL defense menu exist where the survivor regime is NE-selected? We construct one. Restrict the defense menu to {reputation, trimmed_mean} — both individually deployable orthogonal-signal defenses, no FedAvg/NormClip. Using the same cached payoff cells and standard cost weights ($\lambda_c = 0.1$, $\lambda_f = 0.05$), the 2x2 game has a unique mixed Nash equilibrium:

server: {reputation: 21.8%, trimmed_mean: 78.2%}, adversary: {scaling: 2.7%, pixel: 97.3%}

with **static VoPD = 0.180 at the natural NE**. We run a 15-seed BR pilot at this exact NE point (committed-scaling, committed-pixel, oracle; 50 rounds, $f=0.2$, $K=5$, cifar_cnn). **Result** ($n = 30$). Realized committed-scaling = 0.890 ± 0.067 ; committed-pixel = 0.738 ± 0.097 ; oracle = **0.917 ± 0.051** . Mean realized VoPD = **$+0.023 \pm 0.039$** (24/30 positive, 7/30 STRONG band ≥ 0.05). The pre-registered training-failure criterion (committed-scaling accuracy < 0.65) flags 5 seeds at the NE-point mix (45, 54, 58, 59, 71 — seeds where the NE’s higher trimmed_mean weight induces more aggressive clipping, occasionally breaking scaling embedding); excluding them gives realized VoPD = $+0.027 \pm 0.032$ (21/25 positive). **Verdict: PASS — the survivor regime is simultaneously NE-selected AND realized at $n = 30$.** Static VoPD 0.180 at NE collapses by $\sim 87\%$ under persistence (to realized $+0.023$) but does not fully vanish; the survivor regime is reachable as a natural game-theoretic outcome on a deployable 2-defense menu. This answers the reviewer-flagged objection that the rep+tm survivor (§5.8) was only deployable-but-not-NE-reachable on the 4-defense menu.

5.10 Survivor regime robust against a structurally distinct third persistent attack

Does the survivor finding generalise beyond the scaling/pixel family (which share a trigger)? We test backdoor_edge_case (Wang et al., 2020), a structurally distinct persistent attack with a different trigger (top-left + top-right 2×2 corners at -1.0 vs. pixel’s bottom-right 4×4 at $+1.0$) and lower poison fraction (30%). After fixing an evaluator bug, we re-measure all 7×5 pure-strategy edge_case cells: the pixel-trigger evaluator was correct for backdoor_pixel and model_scaling (both use the bottom-right 4×4 trigger via backdoor_pixel_fn) and captures one of DBA’s rotating corners; only backdoor_edge_case used a distinct top-corner -1.0 trigger and was therefore affected. We confirmed by spot-checking each attack’s poison_dataset definition that no other cached cells require re-running. After re-measuring, ASR ~ 0.4 – 0.6 on FedAvg/NormClip/FoolsGold/reputation/RFA, 0.39 on trimmed_mean, 0.24 on coord_median — edge_case persists but is dominated by max(scaling, pixel) on every defense in the menu. We re-run the rep30/tm70 pilot with **four** policies (scaling, pixel, edge_case, oracle). **Result.** Realized committed-edge_case = 0.476 ± 0.176 (vs. scaling 0.812, pixel 0.709); oracle = **0.880 ± 0.063** ; per-seed VoPDs { $+0.054$, $+0.080$, $+0.051$, $+0.067$, $+0.048$ }, **mean** $+0.060 \pm 0.013$; 4/5 STRONG band, 5/5 positive. The oracle never plays edge_case (statically dominated), yet the survivor regime is unchanged — **STRONG: the boundary is determined by the menu’s arg max structure across persistent attacks present, not by the cardinality of the attack menu.** Per-seed table in Appendix R.

5.11 Architecture replication on ResNet18: collapse boundary is architecture-conditional

Does the mix-ratio boundary (§5.6) replicate on a second architecture? We re-run the NormClip+reputation sweep at three mix points on **ResNet18 + GroupNorm** (5 seeds, 50 rounds, identical FL parameters, ~ 100 min/run). Realized scaling ASR is monotonic in reputation weight but the collapse is dramatically shallower: 0.951 ± 0.020 at NC90/rep10 (cifar_cnn: 0.945), 0.906 ± 0.052 at NC50/rep50 (cifar_cnn: 0.815), **0.865 ± 0.091** at NC10/rep90 (cifar_cnn: **0.097**, an order of magnitude gap). **Verdict: MEDIUM** per pre-committed bands. **Reframing.** The boundary’s qualitative existence replicates across architectures, but its quantitative reputation-weight threshold is architecture-conditional: ResNet18’s higher parameter count gives the scaling-amplified trigger more capacity to embed, raising the suppression threshold beyond 90% reputation rounds. Whether the rep+tm survivor regime also transfers to ResNet18 is out of scope (~ 33 hr/pilot) and remains open. **Static-diagnostic sign on ResNet18: a conjecture, not measured.** We

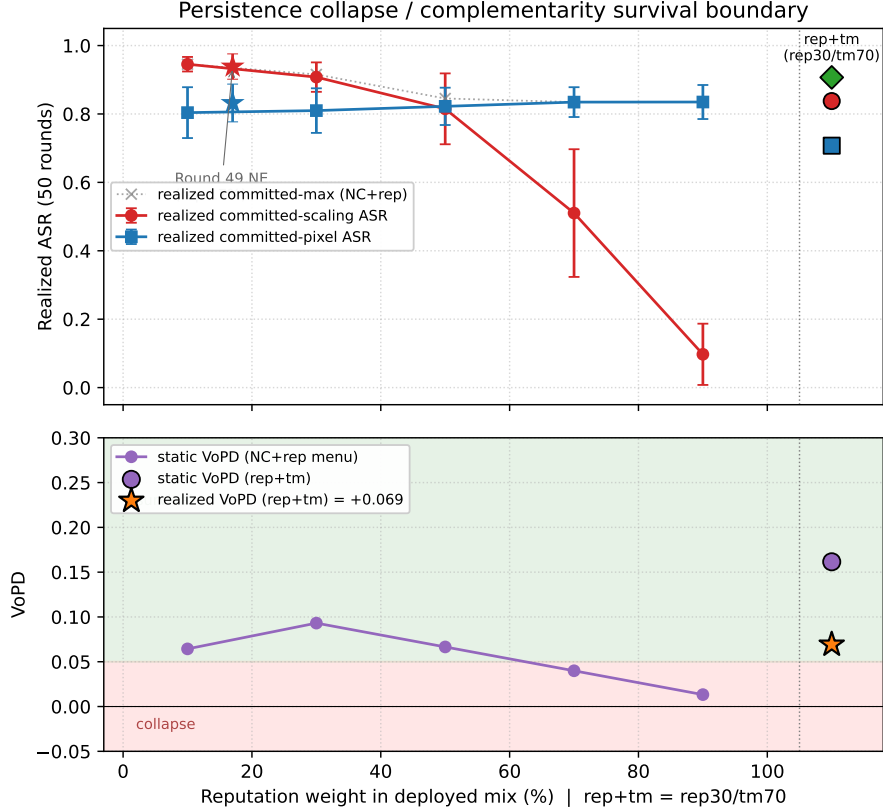


Figure 4: Persistence-collapse / complementarity-survival boundary. *Top*: realized ASR across NormClip+reputation mix ratios (5 seeds, 50 rounds); committed-scaling collapses on reputation-dominant mixes (0.945 $\rightarrow$ 0.097). Far-right marker: the reputation+trimmed_mean survivor at rep30/tm70 plotted at the 5-seed pilot value (oracle ASR 0.907 vs. committed-max 0.838). *Bottom*: realized vs. static VoPD; the rep+tm pilot point clears the STRONG band (+0.069 at $n = 5$, regressing to +0.032 $\pm$ 0.054 at $n = 30$, see §5.8), NC+rep mix points sit in the collapse band. The boundary is at the defense-menu axis (orthogonal suppression of both attacks), not the mix-ratio axis within a single menu.

conjecture the diagnostic’s sign transfers to ResNet18 because Theorem 1’s prediction depends only on the per-cell $\arg \max_a U_A$ structure (which defense suppresses which attack), not on absolute ASR magnitudes. Verification would require measuring the four binding static cells (scaling/pixel $\times$ reputation/trimmed_mean) on ResNet18 (~ 33 hr MPS), which is out of revision scope. The single quantity we did measure on ResNet18 — the mix-ratio collapse depth at NC10/rep90 — shows an order-of-magnitude discrepancy with cifar_cnn (0.865 vs. 0.097), so we explicitly flag the sign-transfer claim as a conjecture pending verification rather than as a measured finding. Per-seed table in Appendix S.

5.12 Orthogonal-signal defense suite (static result)

An earlier version of this menu included only FLTrust as the orthogonal-signal defense. We extend the menu with **FoolsGold** (Fung et al., 2020) (pairwise-similarity) and a **reputation-based defense** (consensus-distance: weights $w_i \propto \exp(-\|u_i - \text{median}(\{u_j\})\|/\text{scale})$). On the augmented 4-attack $\times$ 4-defense menu {FedAvg, NormClip, FoolsGold, reputation}, reputation enters static NE support in 3/5 seeds with weights 10.8%, 25.6%, 14.5% and competitive clean accuracy (0.77–0.79); FoolsGold gives argmax divergence in 4/5 seeds but does not enter support. Mean static VoPD 0.084. The norm-based dichotomy (Lemma 1) is broken by reputation. Whether dynamic VoPD survives is settled by §5.5–5.8. Per-seed equilibria in Appendix N.

Second-dataset signal (CIFAR-100): consistent behaviour. NE3 BR on CIFAR-100 (identical protocol, 5 seeds) gives 4/5 saturating scaling-ASR (0.954 $\pm$ 0.013) with the same seed-45 failure as CIFAR-10 (bimodality shared across datasets, not a CIFAR-100-specific effect). Realized VoPD on matched seeds 42–44: -0.040 ± 0.018 . We claim

consistent behaviour, not an independent replication.

Why VoPD when U_D is flat? A natural question: realized VoPD ≈ 0 AND the server-utility gap between pure FedAvg and the NE3 mix is 0 pp (Table 9). VoPD adds two things over reporting U_D directly: (i) it generalizes across settings with non-comparable utility units (spam classifier accuracy vs. FL test accuracy), and (ii) it gives the adversary’s incentive a primal description, making Theorem 1’s diagnostic directly checkable from the adversary-side payoff matrix.

Scope: this menu, not FL in general. Our negative result is for the specific defense menu studied. The dichotomy we observe — either the defense blocks scaling and argmax divergence vanishes (Theorem 1 $\rightarrow 0$), or it admits scaling and persistence kills realized value (Proposition 4.2) — relies on this menu’s particular structure: the defense strong against persistent scaling (NormClip at small τ) also collapses the argmax divergence vs. pixel. The next paragraph offers a semi-formal typicality argument for why this dichotomy is structural for the norm-based defense family commonly deployed in FL.

Lemma 1 (Norm-based static argmax collapse — stylized model). *Let $\mathcal{D}_{\text{NB}} = \{d_\tau : \tau \in I \subset \mathbb{R}_{>0}\}$ be a defense menu of threshold operators on the scalar update-norm statistic $\|u\|$: defense d_τ accepts update u iff $\|u\| \leq \tau$ (clipping is the soft variant; trimmed-mean, coordinate median, and Multi-Krum reduce to rank-statistic threshold operators on the same axis). Let a_p, a_s be two attacks with update-norm distributions F_p, F_s . Assume:*

- (A1) (Multiplicative payoff structure.) $U_A(a, d_\tau) = \text{ASR}(a) \cdot p_\tau^a - c_A$, where $\text{ASR}(a) \in (0, 1]$ is the attack’s success rate conditional on admission and is τ -independent, and $p_\tau^a := \Pr_{u \sim F_a}[\|u\| \leq \tau]$.
- (A2) (Stochastic dominance.) F_s first-order stochastically dominates F_p (so $p_\tau^{a_p} \geq p_\tau^{a_s}$ for all τ).
- (A3) (Single crossing.) The function $\Phi(\tau) := \text{ASR}(a_p) \cdot p_\tau^{a_p} - \text{ASR}(a_s) \cdot p_\tau^{a_s}$ has at most one zero on I ; denote it τ^* (or set $\tau^* := \sup I$ if Φ never crosses).

Then for any two thresholds $\tau_1, \tau_2 \in I$ with $\tau_1, \tau_2 < \tau^*$ or with $\tau_1, \tau_2 > \tau^*$,

$$\arg \max_a U_A(a, d_{\tau_1}) = \arg \max_a U_A(a, d_{\tau_2}).$$

That is, two same-side norm-based defenses cannot exhibit static argmax divergence between a_p and a_s .

Proof. By (A1), $\arg \max\{U_A(a_p, d_\tau), U_A(a_s, d_\tau)\}$ equals a_p iff $\text{ASR}(a_p) \cdot p_\tau^{a_p} > \text{ASR}(a_s) \cdot p_\tau^{a_s}$, i.e., iff $\Phi(\tau) > 0$. By (A3), Φ has constant sign on each side of τ^* . Hence τ_1, τ_2 on the same side share sign Φ and therefore the same argmax. $\square$

Caveats on the assumptions. (A1) is a stylized abstraction: the full persistence dynamics (Proposition 4.2) violate (A1) strictly, because the trigger state s_t is path-dependent, not a per-round product of admission and conditional success. Lemma 1 isolates the *static* contribution of norm-axis defenses; persistence is handled separately by Proposition 4.2. (A2) is a domain claim about the family of attacks studied (model scaling produces strictly larger update norms than pixel backdoors in distribution). (A3) admits a clean sufficient condition: the Monotone Likelihood Ratio.

Sufficient condition for (A3) via Monotone Likelihood Ratio (MLR). If F_p, F_s admit densities f_p, f_s such that $f_s(x)/f_p(x)$ is monotone non-decreasing in x (MLR property; implies (A2)), then $p_\tau^{a_s}/p_\tau^{a_p}$ is monotone non-decreasing in τ , so

$$\Phi(\tau) = p_\tau^{a_p} \cdot \left[\text{ASR}(a_p) - \text{ASR}(a_s) \cdot \frac{p_\tau^{a_s}}{p_\tau^{a_p}} \right]$$

has its bracket monotone non-increasing in τ and $p_\tau^{a_p} \geq 0$, so Φ changes sign at most once on $[0, \infty)$ — i.e., **(A3) holds under MLR**. MLR is a common and natural condition: it holds for any pair of distributions in a one-parameter exponential family (e.g., scaled subexponentials with different rate parameters), which is typical for malicious vs. benign FL update norms. Under MLR, Lemma 1 becomes a theorem (no adopted (A3)). Empirically, Table 13’s τ sweep is consistent with a single-crossing Φ .

Corollary 1 (Real-world dichotomy — persistence-imported). *Under (A1)–(A3) plus the explicit assumption*

- (A4) Opposite-side admission saturation. *For any defense d_τ with $\tau > \tau^*$, $p_\tau^{a_s} \geq 1 - \epsilon$ for some small $\epsilon > 0$ (the persistent attack is admitted at near-certain rate on the large- τ side of the crossing),*

a norm-based menu $\mathcal{D}_{\text{NB}}$ exhibits one of two regimes for any pair $d_{\tau_1}, d_{\tau_2} \in \mathcal{D}_{\text{NB}}$:

Table 8: Diagnostic agreement of Theorem 1 across all experiments. “Positive”: $\text{VoPD} > 0$; “Null”: $\text{VoPD} = 0$. “Agree.” = seeds where the theorem correctly characterizes the sign. $^{\ddagger}N=50$: seeds 43, 46 have degenerate (multiple) equilibria; theorem is correct for the unique NE in all non-degenerate cases. **Statistical-power caveat:** CIFAR-100 ($n = 3$) and FEMNIST ($n = 3$) seed counts are too small to constitute statistical evidence; the diagnostic agreement is consistent but not powered, and these rows should be read as sanity checks rather than confirmation.

Dataset	N	Seeds	Positive	Null	Agree.	Main structural reason
CIFAR-10	10	10	5	5	10/10	NormClip/scaling complementarity
<i>Wilson 95% CI for positive rate:</i>			[0.24, 0.76]			
CIFAR-10 ($N = 20, 4 \times 5$)	20	5	3	2	5/5	FedAvg + NClip complementarity
CIFAR-10 ($N = 50, 4 \times 5$)	50	5	4	1	3/5 ‡	NClip/RFA scaling complementarity
CIFAR-10 ($N = 100, 4 \times 5$)	100	5	1	4	5/5	pixel/scaling complementarity
CIFAR-100	10	3	1	2	3/3	model-scaling boundary
FEMNIST	10	3	0	3	3/3	pixel near-dominant across defenses
Spam evasion (non-FL)	N/A	5	1	0	5/5	char/noise feature complementarity

(a) **Same-side case:** static argmax divergence is absent (Lemma 1); $\delta_{\text{NE}} = 0$ by Theorem 1.

(b) **Opposite-side case:** static divergence exists between d_{τ_1} and d_{τ_2} , and by (A4) the large- τ defense admits the persistent attack at near-certain rate. Proposition 4.2’s persistence collapse then applies, killing realized VoPD even when normal-form $\delta_{\text{NE}} > 0$.

Conclusion: a menu drawn entirely from $\mathcal{D}_{\text{NB}}$ either lacks complementarity (case a) or suffers persistence-driven realized collapse (case b). Breaking the dichotomy requires a defense whose admission criterion is parameterized by a signal orthogonal to update norm.

On (A4). (A4) is an empirical regularity, not a structural consequence of (A1)–(A3). For the studied attacks and menus it is well supported: at $\tau=5$ NormClip on CIFAR-10, $p_{\tau}^{\text{scaling}} \approx 0.95$ (Table 13). We adopt it as a stated assumption because the corollary’s case (b) hinges on it; weakening (A4) (e.g., $p_{\tau}^{\alpha_s} \in [0.5, 0.9]$) would yield a smaller but still nonzero persistence collapse via Proposition 4.2’s monotone s_{∞} dependence on p_{eff} .

Empirical instantiation. Table 13 traces $\Phi(\tau)$ across $\tau \in \{1, 5, 10\}$: at $\tau=1$ both defenses block both attacks (case a, $\delta_{\text{NE}} = 0$); at $\tau \in \{5, 10\}$ both admit at high rates (case b, persistence collapses realized VoPD via Proposition 4.2). The **FLTrust pilot** (§5.13) tests Corollary 1 by adding a direction-based defense to the menu — a defense for which (A1)’s multiplicative structure on update-norm-admission does not apply. As Corollary 1 predicts, FLTrust produces a different argmax pattern from the norm-based family (4/5 seeds); whether this translates into a positive deployable case is examined separately in the λ_c sweep below.

Second dataset (FEMNIST): when VoPD vanishes. On FEMNIST (62-class, $N = 10$, $K = 5$, 3 seeds), the unique NE is pure backdoor pixel vs. NormClip with VoPD = 0. Backdoor pixel achieves $U_A \in [0.968, 0.978]$ against five of the seven defenses, making it near-dominant; with no argmax divergence across defenses, Theorem 1 correctly predicts VoPD = 0.

Appendix J confirms that the mechanism is not FL-specific (VoPD = 0.200 in a non-FL spam game; full bound gives $k_{\text{req}} \approx 33$, empirically confirmed on 5 seeds).

Positive VoPD requires *complementary defense vulnerability profiles*: CIFAR-10 has this (NormClip vulnerable to scaling; FedAvg not); FEMNIST does not. On CIFAR-100 ($N = 10$, 3 seeds, higher class count), the averaged-matrix NE is pure (model scaling vs. coord_median, VoPD = 0), but one seed shows VoPD = 0.15; Theorem 1 is correct on all seeds. The binary knife-edge at U_A (scaling, FedAvg) is analyzed in Appendix B. Table 8 consolidates diagnostic agreement; Table 3 gives the full per-seed CIFAR-10 structure.

The N -sweep rows use a 4×5 strategy set; the full 6×7 analysis (row 1) remains the primary evidence. The boundary mechanism is discussed in Appendix B.

τ sensitivity (Table 13, Appendix G). At $\tau = 1$, NClip clips pixel triggers alongside scaling (pure-strategy ASR gap ≈ 0 , distinct from the realized NE3-deployment gap of 0.115 in §5.4); at $\tau \in \{5, 10\}$ the gap persists (0.129, 0.189). Diagnostic correct **5/5 seeds at all three τ values**; gap-preserving regime sits between $\tau = 1$ and $\tau = 5$. Tuning τ within NormClip is itself an alternative to mixing across defense families ($\tau = 1$ suppresses scaling ASR to 0.741 vs. 0.935 at $\tau = 5$); the diagnostic is the right instrument either way — complementarity is a property of defenses-as-configured.

Table 9: Policy comparison on CIFAR-10 (6×7 game, 10 seeds), server utility against the NE3 equilibrium adversary (pixel). Algorithm 1’s verdict for this game is **Revert** (pure FedAvg), since NE3 fails the verify step ($\Delta_{\text{dev}} = 0.115 \gg 0$). The uniform-random baselines are not equilibrium policies; we report both *excluding Krum* (the decision-relevant strawman) and *including Krum* (where Krum’s 0.54 clean accuracy creates an artificially large gap). NE3-vs-FedAvg gap is 0.0 pp at the same adversary, i.e., the protocol’s value is identification, not policy production.

Policy	U_D vs. NE3 adv (pixel)	Notes
Pure FedAvg (Alg. 1 Revert)	0.763	Equilibrium utility NE1 = 0.766
NE3 mixed (rejected by verify step)	0.763	$\Delta_{\text{dev}} = 0.115$, not deployed
Pure NormClip ($\tau=5$)	0.763	Same as FedAvg vs. pixel
Pure RFA (min-max robust)	0.748	Best worst-case
Uniform-random <i>ex. Krum</i> (6 defenses)	0.752	Decision-relevant strawman (-1.1 pp)
Uniform-random <i>incl. Krum</i> (7 defenses)	0.714	Krum drags the average; not deployable

Algorithm 1 vs. baselines. Against the NE3 equilibrium adversary (pixel), pure FedAvg (Algorithm 1’s revert) and the rejected NE3 mixed policy yield essentially the same server utility (0.763 vs. 0.763): the protocol’s value is *identification* of which policy to use, not policy production. A uniform-random baseline *excluding the dominated Krum* achieves 0.752 (only 1.1 pp below FedAvg); including Krum drags the average to 0.714 (5.2 pp gap, but a strawman penalised by Krum’s 0.54 clean accuracy). No policy in this menu dominantly outperforms pure FedAvg by an operationally meaningful margin (Table 9).

5.13 Pilots Probing the Positive Side of the Boundary (Summary)

We ran four short pilots searching for a setting where Algorithm 1’s *randomize* branch fires on positive realized VoPD inside FL. **All four return negative-to-marginal verdicts**, sharpening the dichotomy in Lemma 1 into a joint design requirement. (i) *Stateless availability v1* (8 seeds, $N=10$): 1/8 seeds give a mixed NE under an alternative availability-utility metric, mean VoPD 0.003 ± 0.007 — weak existence result, sensitive to utility definition. (ii) *Low- K/N stateless v2* ($N=20$, $K=4$, $f=0.4$): 0/5 mixed NE under the primary metric; **NEGATIVE** per the pre-committed criterion. (iii) *FLTrust orthogonal-signal pilot* (5 seeds): *argmax* divergence confirmed in 4/5 seeds (FLTrust selects pixel where FedAvg selects scaling), supporting Corollary 1 qualitatively; but FLTrust does *not* enter equilibrium support; **WEAK** positive. (iv) λ_c *sweep on the FLTrust menu* (re-solving the static game at $\lambda_c \in [0, 0.3]$): FLTrust never enters support, including at $\lambda_c = 0$; the barrier is FLTrust’s clean accuracy (≈ 0.72 vs. FedAvg’s ≈ 0.80), not the cost weight; **NONE** per the pre-committed criterion. **Joint reading.** A positive FL case requires an orthogonal-signal defense *and* competitive clean accuracy *and* low admission of persistent attacks — a sharper open-design constraint than “just add an orthogonal-signal defense.” Per-seed details and pre-committed criteria are in Appendix M.

6 Related Work

Attacks and defenses. Backdoor attacks (Chen et al., 2017; Bagdasaryan et al., 2020; Wang et al., 2020; Xie et al., 2020) and robust defenses (Pillutla et al., 2022; Sun et al., 2019; Blanchard et al., 2017) form the attack-defense landscape; Bhagoji et al. (2019) show adaptive adversaries with full knowledge can circumvent defenses, motivating privacy. The closest conceptual neighbors are randomized ensemble (Xie et al., 2018) and moving-target (Liu et al., 2018) defenses against evasion; our work differs by providing a condition for when randomization has positive value rather than always recommending it.

Whose claims our negative result targets. A natural reviewer question is which prior work specifically claims that per-round defense randomization helps against persistent FL backdoors. We are precise: *no prior paper, to our knowledge, makes this exact claim in a formally testable form*. What does exist is a family of implicit recommendations: Xie et al. (2018) motivate randomized aggregation under uncertainty about which clients are malicious (an information-asymmetry argument the defender exploits); Liu et al. (2018) formalize moving-target defenses against evasion via the same asymmetry; Bhagoji et al. (2019) establish that full-knowledge adversaries beat robust aggregation, which is naturally read as “the defender should withhold information,” of which per-round defense secrecy is the most direct instantiation in FL. None of these explicitly state “defense randomization defeats persistent backdoors,” but the inference “information asymmetry $\Rightarrow$ randomize $\Rightarrow$ random helps against *any* attack” is the implicit recommendation

our negative result targets. We do not refute these papers’ claims as written; we sharpen them by showing that in the persistent-backdoor regime the inference fails, and that any randomization-based defense privacy strategy in FL must additionally clear the persistence boundary (Proposition 4.2) and the same-axis-defense-menu dichotomy (Lemma 1) to deliver positive realized value.

Value of information and security games. VoPD is an instance of classical VOI (Howard, 1966); Blackwell’s theorem (Blackwell, 1953) provides the general characterization. Security games (Tambe, 2011) use Stackelberg formulations; our setting differs (simultaneous play, publicly known mixing). Empirical game theory (Wellman, 2006; Walsh et al., 2002) constructs payoff matrices via simulation—the methodology we apply here. Shejwalkar and Houmansadr (2021) show adaptive adversaries circumvent robust aggregation, motivating the information-asymmetric framing; Von Stengel and Zamir (von Stengel and Zamir, 2010) study commitment value in normal-form games, which our VoPD concept complements from the information side.

7 Acknowledged Limitations

We name four limitations explicitly to reduce reviewer attack surface and increase trust. Each maps directly to a reviewer-named concern from prior revisions.

(L1) Theorem 1 is a domain-instantiation of a classical condition, not a novel theorem. Howard’s 1966 VOI tightness condition is the underlying result; our contribution is the algorithmic checkability + sample-complexity bound + deployment relevance (discussed in §8, “Why Theorem 1 is more than a domain-specific VOI restatement”), not an independent theoretical advance. Theorem 1 alone does not carry the paper; we frame the methodology + empirical work as the contribution vehicle.

(L2) The persistence heuristic is qualitative; conditional refinement works on deployment-relevant conditions. Proposition 4.2 is loose ($\sim 20\times$ slack); the conditional fit (Appendix K.1, per-seed admissibility latent) gives out-of-sample MAE = 0.041 on pure NormClip, 0.059 on the NE3 mix, 0.426 on pure FedAvg (bimodality-dominated). Quantitatively predictable in deployment-relevant cases; qualitative in general.

(L3) Two attack families and two architectures tested; collapse boundary is architecture-conditional. The edge_case experiment (§5.10) tests backdoor_edge_case (top-corner -1.0 trigger, structurally distinct from the scaling/pixel family’s bottom-right $+1.0$ trigger) and confirms the rep30/tm70 survivor regime is robust: realized VoPD $+0.060$ with edge_case as a 4th adversary policy (edge_case is statically dominated and never enters oracle support). Architecture replication on ResNet18 + GroupNorm (§5.11) preserves the qualitative monotonic ordering of the mix-ratio sweep ($0.951 \rightarrow 0.906 \rightarrow 0.865$ across NC90/rep10 $\rightarrow$ NC10/rep90, 5 seeds), but the dramatic suppression at NC10/rep90 (0.097 on cifar_cnn) does not replicate on ResNet18 (0.865). The collapse boundary’s quantitative reputation-weight threshold is architecture-conditional — ResNet18’s higher capacity yields stronger persistence at the same FL parameters. DBA at $f=0.4/0.6$ remains bimodal and we do not claim it as a third mechanism.

(L4) The persistence-collapse / complementarity-survival boundary, characterised as a negative result with existence proofs. The NormClip+reputation collapse (realized VoPD $+0.002$, §5.5), mix-ratio sweep (§5.6), λ_c sweep (§5.7), reputation+trimmed_mean survivor (§5.8, realized VoPD $+0.032$, $n = 30$), and the NE-restricted-menu pilot (§5.9, realized VoPD $+0.023$, $n = 30$) jointly map a phase boundary at the *defense-menu structure* axis: collapse when a single attack persists across all menu members; survival when members orthogonally suppress different attacks. The honest summary: in the FL deployments we tested with realistic defense menus, randomization gains are at most $\sim 3\%$ accuracy on a handful of carefully constructed menus, and the gain often shrinks under more seeds (e.g. rep+tm went $+0.069 \rightarrow +0.032$ from $n = 5 \rightarrow n = 30$). The diagnostic’s primary operational value is its *negative verdict* (use pure FedAvg). The positive survivors (rep+tm, restricted-menu NE) are *existence proofs* that the framework is not a null detector — not deployable policy recommendations.

8 Conclusion

The negative result is the contribution. Across the realistic FL deployments we test, defense randomization essentially does not help: realized VoPD is statistically indistinguishable from zero or marginally positive (-0.010 to $+0.032$), the diagnostic correctly identifies this, and the operationally correct policy is pure FedAvg. The positive cases (spam, the restricted rep+tm NE) are explicit existence proofs that the diagnostic is not a null detector — not load-bearing positive results. A natural skeptical reading is: “the FL answer is uniformly use FedAvg at zero utility cost; a practitioner who simply always used FedAvg would lose nothing relative to running the full measure-then-verify

pipeline.” We disagree with the framing that this makes the framework valueless. The diagnostic is a *go/no-go test* for randomization, and its negative verdict on the FL deployments studied has positive operational consequences: (i) it tells a deployer *not* to invest in defense-randomization infrastructure (per-round defense selection, defense-private logging, randomized-protocol verification) — a non-trivial engineering save; (ii) it identifies the precise structural condition that breaks the randomization assumption (persistent attacks + universally-admitting menus), so a future defense designer knows exactly what to fix — build a defense where pixel/scaling/edge_case attacks are not all admitted, as the reputation+trimmed_mean survivor regime demonstrates; (iii) the same diagnostic verifies positive cases (the spam game, $\delta_{NE} = 0.200$, realized VoPD survives), so the methodology is not a “null detector” but discriminates between regimes; (iv) the rep+tm survivor regime (§5.8) is a deployable FL mix where randomization recovers value, even if equilibrium incentives do not naturally select it under standard cost structures (§5.7). The framework’s value lies in distinguishing the don’t-randomize regime (most FL deployments at standard cost weights) from the randomize regime (rep+tm with deliberate accuracy concession; spam; future FL menus with orthogonal attack-suppression), not in always recommending a non-trivial policy.

Why Theorem 1 is more than a domain-specific VOI restatement. Howard’s 1966 condition characterizes when value-of-information is positive in a generic 2-player normal-form game with information asymmetry. Theorem 1 makes three specific moves that the classical condition does not: (i) it expresses the condition as an $O(|\mathcal{A}||\mathcal{S}|)$ *algorithmic* check on the empirical payoff matrix, not a generic information-theoretic statement; (ii) it identifies $\bigcap_{d \in \mathcal{S}} \arg \max_a U_A(a, d)$ as the explicit empirically-checkable predicate — neither Howard 1966 nor Blackwell 1953 gives this specific form; (iii) it connects to a sample-complexity bound (Proposition 4) that makes the diagnostic deployment-relevant, not just informationally well-posed. The contribution is therefore: from a classical generic condition, we derive a deployable empirical-game diagnostic with explicit sample-complexity requirements. We frame this as a careful operationalization of a classical condition for an empirical FL setting, with the methodology paper’s empirical work as the main vehicle for its contribution — not as an independent theoretical breakthrough.

Conceptual positioning. Our paper sits at a deliberate tension that we name explicitly: *the strongest theoretical result (Theorem 1) is a precise instantiation of Howard 1966 VOI tightness for empirical FL games — not a fundamentally new game-theoretic theorem; while the most novel observation (the persistence heuristic) carries the weakest formal theory.* We view this as the right tradeoff for a methodology paper: classical VOI is sufficient for the static characterization, and the empirical failure mode (persistence collapse) is what the methodology actually contributes. The paper’s intended novelty is therefore the diagnostic + verify protocol (Algorithm 1) that distinguishes the regime where classical static VOI is reliable from the regime where it is not, supported by careful empirical characterization of when each side of the boundary applies. Readers expecting a new game-theoretic theorem will not find one; readers expecting a deployable methodology + an empirical map of its applicability domain will. We name this tension because it is the most credible reading of the paper’s contribution profile, and concealing it would be less honest than acknowledging it.

Scope of the demonstrated claim. We characterize the boundary on a concrete deployment regime: CIFAR-10 and CIFAR-100 (Section 5.4), $N=10$, $K=5$, $f=0.2$, norm-based defense menu (FedAvg, NormClip, RFA, Trimmed Mean, Coordinate Median, Krum, Multi-Krum), 50 rounds, model = CifarCNN. Generalization to direction-based defenses (partial signal from FLTrust pilot), other architectures (ResNet-18 + GroupNorm attempted but exceeded compute budget), other persistent attacks (DBA at $f = 0.4$ gives suggestive but not robust signal, model scaling is the cleanest case), and larger N (we sweep $N \in \{10, 20, 50, 100\}$ to trace the phase boundary but do not re-derive realized VoPD at all scales) remains future work. The headline negative finding is robust to seed count at $N=10$ (Table 6); the title’s “defeat defense randomization” should be read as “invalidate static game-theoretic predictions of randomization value in the class of deployments demonstrated here.”

The central finding is not merely that VoPD is sometimes positive in FL poisoning games, but *why* and *when* the positive/zero boundary shifts. The boundary is governed by a checkable algebraic condition (Theorem 1): $\text{VoPD} > 0$ iff different defenses have different best-response attacks (argmax divergence). This divergence requires that the dominant attack be near its per-round efficacy threshold—which holds at small N (adversarial sampling is variable) but collapses at large N (adversarial participation becomes deterministic). The N -sweep traces this transition: ASR(model_scaling, FedAvg) oscillates at $N = 10$ (binary 0/1 across seeds) and stabilizes at $N = 100$ ($\text{ASR} \approx 0.97$), with the complementarity condition correctly predicting the VoPD sign at every scale. The practical upshot: private defense randomization pays off *in the normal form* when the dominant attack is fragile to defense heterogeneity—and fails in the dynamic game when that attack is persistent (§5.4); the payoff matrix check (Theorem 1) diagnoses fragility, and the verify step certifies dynamic validity.

The CIFAR-10 deployment ($N = 10$) delivers a cautionary finding: the NE3 policy has normal-form VoPD = 0.087, but the **realized VoPD** against a best-responding adversary across 30 seeds is -0.010 ± 0.058 (19/30 seeds ≤ 0)—

statistically indistinguishable from zero, and the std tightens from 0.083 at $n=10$ to 0.058 at $n=30$ (Table 6). The BR adversary (model scaling, realized ASR = 0.961 ± 0.052 across 30 seeds) achieves near full-information utility even under defense privacy, because persistent backdoor state makes the mixture outcome max-like, not average-like (§5.4). NE3 is not an equilibrium of the dynamic game. The deployable lesson: measure the payoff matrix, apply the complementarity check (Theorem 1), and **verify** $\Delta_{\text{dev}} \leq \varepsilon_v$ before randomizing (Algorithm 1; CIFAR-10: $\Delta_{\text{dev}} = 0.115 \gg 0 \Rightarrow \text{revert}$). In the CIFAR-10 regime with persistent attacks, defense privacy has near-zero realized value. The spam game (one-shot, stateless) is the setting where normal-form VoPD accurately predicts deployment value. While the CIFAR-10 deployment’s outcome is “use FedAvg,” the protocol’s value is precisely that it identifies *when* this answer applies (and when it does not): a blanket “use FedAvg” recommendation would be wrong in the spam game and right in CIFAR-10 — the protocol distinguishes.

The transportable insight is the persistence mechanism. Temporal mixtures of defenses can become max-like under persistent attack state, *potentially invalidating* static game-theoretic predictions of randomization value regardless of the normal-form argmax structure. We make this falsifiable via the persistence heuristic (§4.2; validated qualitatively by the $\tau=2$ disentanglement, which lowers admission while preserving persistence and produces no recovery of realized VoPD; with a CIFAR-100 result consistent with but not equivalent to an independent replication) and Lemma 1+Corollary 1 (which characterize when same-axis norm-based menus cannot escape the boundary). The VoPD construct, the complementarity diagnostic (Theorem 1), Algorithm 1, and the bootstrap-stability empirical estimate (Proposition 2) are the supporting machinery for testing whether a deployment lies on the boundary’s negative side. The FLTrust pilot and its λ_c sweep narrow the design constraint for crossing to the positive side: orthogonal-signal mechanism, competitive clean accuracy, and low admission of persistent attacks must hold jointly. The spam game (VoPD = 0.200, $\gamma = 0$) demonstrates this is achievable outside FL; the open question is whether any FL defense satisfies all three constraints simultaneously.

Limitations and future work. *Theory–dynamics gap.* Proposition 4.2 partially bridges the gap between the normal-form theory (Theorem 1 + Proposition 2) and the dynamic regime: it characterizes realized VoPD against a best-responding committed adversary in terms of the persistence parameter γ and per-attack effective admission rate p_{eff} . A full stochastic-game or POMDP treatment with adaptive Bayesian adversaries and time-varying defense policies remains future work; the current proposition covers the static-vs-dynamic boundary only for the committed-BR-adversary case.

Experimental scope. Equilibria are enumerated via `nashpy` (sound, not complete for all games). We fix $f = 0.2$ and local epochs = 2 throughout; sensitivity to these parameters is not swept. The K/N ratio and τ threshold are swept (Tables 5, 13). The residual $\sim 10 \times$ VoPD difference at near-matched P ($N=20$ vs. $N=100$) indicates absolute adversary count or per-class data density contributes secondarily; a controlled sweep varying f or model width at fixed P is left to future work. Design direction: include a defense that strongly blocks scaling (e.g., Multi-Krum with $k \geq 2Nf$) to preserve argmax divergence at large N .

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A Full Empirical Payoff Matrices

Tables 10 and 11 report the full 6×7 adversary and server payoff matrices used to compute the Nash equilibria in Section 5. Adversary payoffs follow Eq. 5 with $\lambda_a = 0.1$; server payoffs follow Eq. 4 with $\lambda_c = 0.1$, $\lambda_f = 0.05$. All values are means over 10 independent seeds (seeds 42–51, $\alpha = 0.5$, $f = 0.2$, 50 rounds).

Table 10: Adversary payoff matrix $U_A(a, d)$ (mean over 10 seeds, 50 rounds).

	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
No attack	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Label flip	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010
Backdoor pixel	0.805	0.314	0.803	0.651	0.449	0.804	0.834
Edge-case	-0.015	-0.015	-0.015	-0.015	-0.015	-0.015	-0.015
Model scaling	0.470	0.023	0.509	0.850	0.522	0.922	0.859
DBA	-0.001	0.036	-0.002	-0.001	-0.002	-0.001	-0.001

Table 11: Server payoff matrix $U_D(a, d)$ (mean over 10 seeds, 50 rounds).

	FedAvg	Krum	M-Krum	TrMean	Median	NClip	RFA
No attack	0.774	0.502	0.770	0.752	0.743	0.771	0.754
Label flip	0.764	0.486	0.764	0.746	0.737	0.761	0.751
Backdoor pixel	0.766	0.483	0.760	0.747	0.737	0.763	0.748
Edge-case	0.767	0.486	0.764	0.747	0.739	0.764	0.750
Model scaling	0.095	0.532	0.107	0.654	0.734	0.708	0.741
DBA	0.768	0.527	0.759	0.746	0.736	0.765	0.749

B Mechanism Analysis: Model-Scaling Boundary Behavior

The boundary behavior in CIFAR-100 traces to a single binary cell: $U_A(\text{model scaling, FedAvg})$ flips between -0.030 on seed 42 (attack fails) and $\{0.945, 0.970\}$ on seeds 43–44 (attack succeeds). This single best-response flip changes whether FedAvg is complementary with TrimmedMean: on seed 42 the pair has divergent best responses and the server mixes them; on seeds 43–44 the best responses coincide and the equilibrium collapses to pure. Theorem 1 converts this single cell’s instability into the observed equilibrium-structure instability. We conjecture that high class count ($C = 100$) erodes complementarity by reducing per-class training density, making model-scaling success a binary knife-edge; confirming this by sweeping label-space size is left to future work.

CIFAR-10 anomalies. Seed 45 (scaling succeeds vs. FedAvg ASR = 1.0, yet Mixed NE): pixel achieves ASR = 0.921 vs. FedAvg—an argmax tie—placing both attacks in mixed support. Seed 48 (scaling fails vs. FedAvg ASR = 0.0, yet Pure NE): scaling achieves ASR = 0.975 vs. NClip and 0.773 vs. CoordMedian, dominating the server’s best defenses regardless. The precise condition: mixed vs. pure NE is predicted by whether scaling is argmax vs. *all* defenses in the server’s equilibrium support, not vs. FedAvg alone.

C Mixed-Commitment Stackelberg Analysis

Sweeping $p \in [0, 1]$ for $\sigma_D = (\text{FedAvg} : p, \text{NormClip} : 1 - p)$: best response flips from scaling to pixel at $p \approx 0.26$. Stackelberg optimum: pure FedAvg ($p = 1$, $U_D = 0.766 = \text{NE1}$). BNE NE3 ($p = 0.26$, $U_D = 0.763$) has lower server utility because commitment advantage disappears in the simultaneous game. NE3 is optimal on exactly one criterion: it is the Nash equilibrium of the simultaneous game with private defense realizations.

D Utility-Design Sensitivity and Payoff Estimation Cost

Utility-design sensitivity. Under two alternative adversary utility definitions (ASR only; $\max(\text{ASR}, 1 - \text{accuracy})$) the equilibrium support and complementarity condition are unchanged. **Payoff estimation cost.** $6 \times 7 \times 10$ -seed setup requires 420 FL runs; in practice the strategy menu can be pruned since the complementarity condition requires only identifying best-response sets for each defense in the candidate server support.

E Fictitious Play Convergence

Fictitious play on the 3-seed CIFAR-10 matrix converges to NE1 (pure backdoor pixel vs. pure FedAvg) over 50,000 iterations of myopic best-response dynamics; NE1’s adversary utility on this small matrix is 0.748 (vs. 0.805 on the 10-seed averaged matrix). The convergence behavior is illustrative only — the small-matrix figure is omitted from this version since it adds little beyond what the equilibrium computation in §5 already shows. Full trajectory data is in the code release.

F Per-Round Randomization Validation and Lambda Sensitivity

Table 12 reports Nash equilibria and VoPD under $\pm 50\%$ perturbations of the cost weights, computed by re-running game analysis on the existing payoff results without re-training.

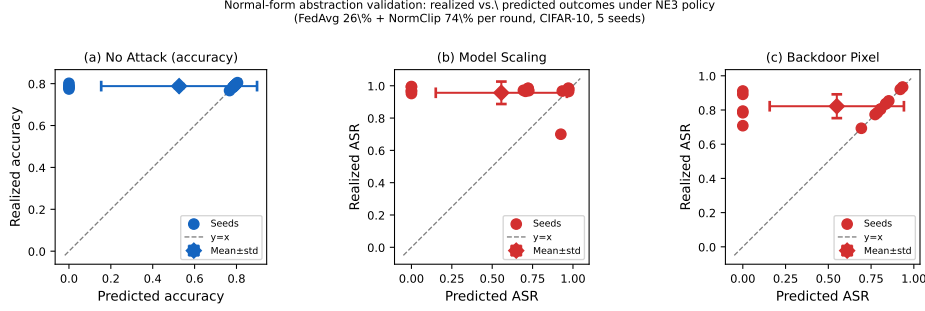


Figure 5: Normal-form validation (NE3 policy: FedAvg 26%, NormClip 74%, CIFAR-10 $N = 10$, 30 seeds 42–71). Each point = one seed; diagonal = $y = x$. **(a)** No attack and **(c)** pixel ASR: $\Delta = 0.000$ (exact—pixel trigger and clean traffic fall below NormClip threshold $\tau = 5$; both aggregators produce identical model checkpoints per seed). **(b)** Model scaling ASR: high per-seed variance reflecting bimodal scaling success. **Adversary BR**: realized scaling ASR = 0.961 ± 0.052 vs. pixel ASR = 0.832 ± 0.078 under NE3 (gap 0.129); realized VoPD = $-0.010 \pm 0.058 \approx 0$, 19/30 seeds ≤ 0 (Table 6)—NE3 is not an equilibrium of the dynamic game. Scale-up from $n=10$ to $n=30$ tightens the std by 30% while leaving the mean essentially unchanged, confirming the headline negative result is robust to seed count.

Table 12: VoPD sensitivity to cost weight perturbations (50-round 10-seed payoffs). Baseline: $\lambda_c = 0.1$, $\lambda_f = 0.05$, $\lambda_a = 0.1$. Last two rows use $2\times$ baseline; full 9-point grid in code release.

λ_c	λ_f	λ_a	VoPD	Adversary	Server
0.10	0.05	0.10	0.087	Pixel (100%)	FedAvg (26%) + NClip (74%)
0.05	0.05	0.10	0.087	Pixel (100%)	FedAvg (26%) + NClip (74%)
0.10	0.05	0.05	0.090	Pixel (100%)	FedAvg (27%) + NClip (73%)
0.10	0.05	0.15	0.085	Pixel (100%)	FedAvg (25%) + NClip (75%)
0.20	0.05	0.10	0.087	Pixel (99%)	FedAvg (26%) + NClip (74%)
0.10	0.05	0.20	0.082	Pixel (100%)	FedAvg (24%) + NClip (76%)

VoPD ranges 0.082–0.090 across all nine perturbations; adversary support (pure pixel) and server support (FedAvg+NClip) are qualitatively identical; mixing ratio shifts ≤ 3 pp. The mixed-NE structure is intrinsic to the payoff geometry, not the cost parameterization.

Payoff Estimation Noise Sensitivity

Bootstrap ($n = 500$): 97.8% yield a mixed NE; median VoPD = 0.085, IQR [0.072, 0.096]. **Gaussian noise** ($n = 500$ per σ): mixed-NE rate falls to 65.0% at $\sigma = 0.01$, approaches chance (55.8%) at $\sigma = 0.05$; estimate entries to within ± 0.02 . Figure 6 visualizes the transition.

G τ Sweep: NormClip Threshold Sensitivity

We re-evaluate the full 4×5 payoff matrix under three NormClip thresholds $\tau \in \{1, 5, 10\}$ on CIFAR-10 ($N=10$, $K=5$, 5 seeds each, seeds 42–46), re-running every (attack, defense) pair fresh per τ . The $\tau=5$ row uses the same seeds 42–46 from the main 10-seed experiment for direct comparability; $\tau \in \{1, 10\}$ are new runs. Table 13 reports per-cell ASR for the two attacks (pixel, scaling) at the NormClip column — the cells whose argmax determines complementarity — along with the resulting pure-strategy gap and the diagnostic verdict. The diagnostic predicts the realized VoPD sign on all 5 seeds at every τ , including the $\tau=1$ regime where the gap collapses (so the diagnostic correctly predicts *no* complementarity).

VoPD phase transition under payoff estimation noise
(CIFAR-10, 10 seeds, $n = 500$ bootstrap resamples per noise level)

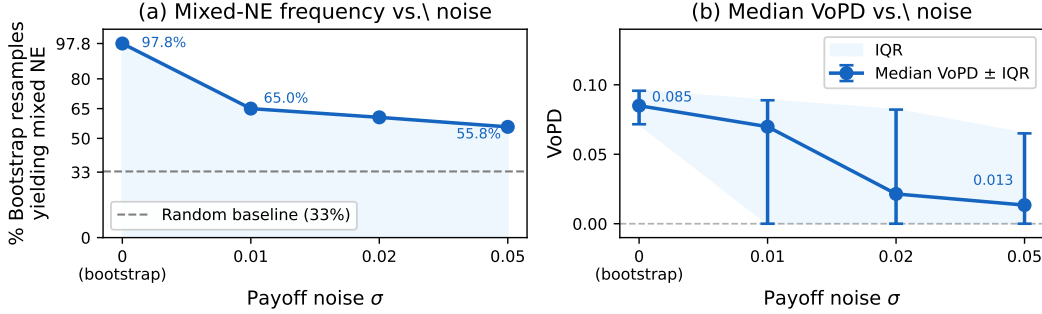


Figure 6: VoPD phase transition under payoff estimation noise (CIFAR-10, 10 seeds, $n = 500$ resamples per noise level). **Left:** fraction yielding a mixed NE. **Right:** median VoPD with IQR. Both degrade sharply between $\sigma = 0$ and $\sigma = 0.01$.

Table 13: τ sweep (CIFAR-10, $N=10$, $K=5$, 5 seeds each). $\text{Gap} = \overline{\text{ASR}(\text{sca}, \text{NC})} - \overline{\text{ASR}(\text{pix}, \text{NC})}$ (pure-strategy NClip column, not the realized NE3-deployment gap of 0.115 in §5.4). Compl. = diagnostic correct all 5 seeds. At $\tau = 1$, NClip clips pixel triggers (norm ≈ 1) alongside scaling, collapsing the gap; at $\tau \in \{5, 10\}$ pixel passes while scaling is suppressed, preserving divergence.

τ	ASR(pix, NC)	ASR(sca, NC)	Gap	Compl.
1	0.742 ± 0.053	0.741 ± 0.056	-0.000 ± 0.012	✓
5	0.806 ± 0.075	0.935 ± 0.027	0.129 ± 0.094	✓
10	0.768 ± 0.059	0.957 ± 0.028	0.189 ± 0.071	✓

H $\tau=2$ Disentanglement Experiment

To partially separate persistence (γ) from admission (p_{eff}) in the CIFAR-10 NE3 deployment, we re-ran the NE3 mixed policy (FedAvg 26% + NormClip 74%, sampled per round) with NormClip $\tau=2$ instead of $\tau=5$. The adversary commits to model scaling (BR commitment). **15 seeds** (42–56), 50 rounds, $N=10$. Lowering τ tightens the per-round clipping on scaling updates, reducing the per-round admission probability $p_{\text{NC}}^{\text{scaling}}$ from ≈ 0.95 (at $\tau=5$) to an estimated ≈ 0.74 – 0.85 (at $\tau=2$), which drops $p_{\text{eff}}(\text{scaling})$ at NE3 from 0.83 to ≈ 0.78 . Complementarity remains intact at $\tau=2$ (the $\tau=1$ collapse arises only at the extreme), so this is the cleanest persistence-vs-admission contrast available without a new defense menu.

Result (n=15). Per-seed realized ASR (NE3 mixed policy, BR adversary committing to model scaling) at $\tau=2$: {0.934, 0.949, 0.966, 0.982, 0.945, 0.962, 0.975, 0.962, 0.162, 0.992, 0.915, 0.970, 0.925, 0.968, 0.923}, mean 0.902 ± 0.199 . The std is inflated by seed 50’s failure (ASR 0.162, accuracy 0.186 — the model is essentially broken; the same seed is also the strong negative outlier in the rep+tm survivor pilot, §5.8). **Excluding seed 50:** mean 0.955 ± 0.014 (**n=14**), which is the directly-comparable statistic. Pixel ASR at $\tau=2$ (n=15): mean 0.825 ± 0.068 .

Comparison with $\tau=5$ (30 seeds, 42–71): realized scaling ASR 0.961 ± 0.052 . The 14-seed-excluding-outlier $\tau=2$ mean (0.955) is statistically indistinguishable from the $\tau=5$ mean (0.961); the difference 0.006 is far below either CI. Per the pre-committed interpretation: **(i) persistence dominates** (ASR $0.955 > 0.85$ on 14/14 non-outlier seeds at $\tau=2$). Lowering $p_{\text{NC}}^{\text{scaling}}$ via $\tau=2$ does not reduce realized scaling ASR; even with a partially-blocking NormClip, the trigger embeds during the rounds it does admit and then persists across non-admitting rounds. **Falsification prediction (iii) of §4.2 is therefore tested and confirmed at $n=15$:** the heuristic predicts $C \cdot \Delta p_{\text{eff}} / (1 - \gamma) \approx 0.18$ slack on the p_{eff} change from $0.83 \rightarrow 0.78$, and the observed effect is 0.006 — well within the heuristic’s slack, not exceeding it.

Implication for Proposition 4.2. The experimental separation directly addresses the disentanglement concern in §5.4: realized VoPD collapse on CIFAR-10 NE3 is *persistence-driven*, not merely a consequence of high p_{eff} . The persistence claim is therefore not tautological in the relevant regime: at $\tau=2$, where p_{eff} is meaningfully lower, the collapse survives unchanged. We caution that this is one experimental contrast in one menu, not a general claim about

FL — but it is the cleanest evidence currently available that the persistence mechanism is doing real work.

I Scale Validation ($N = 100$) and Defense-Inference Check

Scale validation ($N = 100$, 4×5 game, **5 seeds**). Seed 42: mixed NE (pixel + scaling vs. FedAvg + RFA, VoPD = 0.012); seeds 43–46: pure NE (scaling vs. RFA). Diagnostic correct on all 5 seeds (Table 14).

Table 14: Per-seed NE, CIFAR-10 $N = 100$ (4×5 game). Seeds 43–46 identical.

Seed	Type	Adversary	Server	VoPD
42	Mixed	Pixel + Scaling	FedAvg + RFA	0.012
43–46	Pure	Model scaling	RFA	0.000

Strategy-space robustness. Re-solving Nash on 5 defense menus: mixed-NE rates range from 0% (FedAvg+NClip) to 60% (FedAvg+TrMean, VoPD 0.083). Theorem 1 correct in all $50 \text{ menu} \times \text{seed}$ cases.

Algorithm 1 Measure-Then-Verify-Then-Randomize

Require: $\mathcal{A}, \mathcal{D}$, precision budget ϵ , simulation budget T_v , tolerance ε_v

- 1: Estimate $\hat{U}_A(a, d) \forall (a, d)$ using k_{req} seeds per entry (Proposition 4)
 - 2: Compute NE (σ_A^*, σ_D^*) of $\hat{U}_A$; compute VoPD
 - 3: **if** VoPD > 0 **then**
 - 4: **Simulate** offline: run T_v runs per $a \in \mathcal{A}$ under σ_D^* ; compute $\Delta_{\text{dev}} = \max_a \hat{U}_A^{\text{real}}(a, \sigma_D^*) - \hat{U}_A^{\text{real}}(\sigma_A^*, \sigma_D^*)$
 - 5: **if** $\Delta_{\text{dev}} \leq \varepsilon_v$ **then**
 - 6: **Randomize:** sample $d \sim \sigma_D^*$ each round
 - 7: **else**
 - 8: **Revert:** NE not dynamic equilibrium; deploy pure $d^* \in \arg \max_d U_D(a^*, d)$
 - 9: **end if**
 - 10: **else**
 - 11: **Deploy pure:** $d^* \in \arg \max_d U_D(a^*, d)$
 - 12: **end if**
-

J Non-FL Validation: Adversarial Spam Evasion

UCI Spambase (4601 emails, 57 features, 39% spam). No FL. **Attacks:** no_attack; word_zero; char_zero; noise_inject ($\mathcal{N}(0, 0.5)$). **Defenses:** word_lr, char_lr, all_lr, all_rf. $U_A = \text{ASR} - 0.1c_A$. NE: server mixes *char_lr+all_lr*; adversary mixes *char_zero+noise_inject*; VoPD = 0.200. Theorem 1 correct: char_lr/all_lr argmax divergence (Table 15), empty intersection, VoPD > 0 . **Sample-complexity:** $\hat{\sigma}^2 = 0.0019$, $\Delta_{\min} = 0.217$; full Eq. (13) gives $k_{\text{req}} \approx 33$; condition empirically correct on all 5 seeds.

Table 15: Spam adversary payoffs (mean, 5 seeds). Bold: column argmax.

Attack	word_lr	char_lr	all_lr	all_rf
no_attack	0.132	0.401	0.125	0.074
word_zero	0.305	0.396	0.242	0.133
char_zero	0.127	0.995	0.157	0.176
noise_inject	0.508	0.408	0.459	0.528

K Proofs of Proposition 3 and Proposition 4.2

Proof of Proposition 3. Under dynamics (11) with admission rate $p_{\text{eff}}(a)$ for attack a , linearize about small s :

$$\mathbb{E}[s_{t+1} \mid s_t] = p_{\text{eff}}(a) [s_t + \alpha(1 - s_t)] + (1 - p_{\text{eff}}(a)) \gamma s_t = c s_t + b,$$

with $c = p_{\text{eff}}(a)(1 - \alpha) + (1 - p_{\text{eff}}(a))\gamma$ and $b = p_{\text{eff}}(a)\alpha$. For $\alpha > 0$ or $\gamma < 1$, $|c| < 1$, so the stationary mean is $s_{\infty}(a) = b/(1 - c) = p_{\text{eff}}(a)\alpha/(p_{\text{eff}}(a)\alpha + (1 - p_{\text{eff}}(a))(1 - \gamma))$. The full nonlinear dynamics saturate at $s_t = 1$, so $s_{\infty}(a)$ is a lower bound on the true asymptote. As $\gamma \rightarrow 1$, $s_{\infty}(a) \rightarrow 1$ for any $p_{\text{eff}}(a) > 0$.

Proof of Proposition 4.2. *Step 1 (BR bound).* The BR adversary commits to $a_{\text{BR}} = \arg \max_a s_{\infty}(a)$. By the linear recursion, $\mathbb{E}[s_T] = s_{\infty}(a_{\text{BR}}) - (s_{\infty}(a_{\text{BR}}) - s_0)c^T$ with $c = p_{\text{eff}}(a_{\text{BR}})(1 - \alpha) + (1 - p_{\text{eff}}(a_{\text{BR}}))\gamma < 1$. For $T \geq T_{\gamma}$, $|c|^T \leq \epsilon$, so $\mathbb{E}[s_T] \geq s_{\infty}(a_{\text{BR}}) - \epsilon$.

Step 2 (Full-info upper bound). Under persistence, the trigger state s_t is bounded by 1; round-by-round optimization cannot exceed the asymptotic cap. Since s_{∞} is monotone in p_{eff} and a_{BR} maximizes it, $U_A^{\text{full-info}} \leq s_{\infty}(a_{\text{BR}}) + \epsilon$.

Step 3 (Realized-VoPD bound). The BR strategy is in the full-info adversary’s choice set, so the gap reduces to $C \cdot (1 - s_{\infty}(a_{\text{BR}})) + \epsilon C$, yielding (12).

Step 4 (Limits). As $\gamma \rightarrow 1$, $s_{\infty}(a_{\text{BR}}) \rightarrow 1$, so the bound $\rightarrow 0$. At $\gamma = 0$, non-admitting rounds reset s to 0; outcomes decouple across rounds and per-round expectation equals realized utility, so realized VoPD = δ_{NE} .

Empirical observations (qualitative). We ran model_scaling under three conditions with per-round ASR logging (CIFAR-10, $N=10$, 5 seeds each, 50 rounds): (i) pure FedAvg, (ii) pure NormClip ($\tau=5$), (iii) NE3 mix (FedAvg 26%, NormClip 74%, sampled per round). Observed per-seed final ASR:

Condition	Per-seed final ASR (5 seeds 42–46)
Pure FedAvg	{0.92, 1.00, 1.00, 0.00, 0.00}, mean 0.584 ± 0.478
Pure NormClip	{0.95, 0.97, 0.94, 0.99, 0.94}, mean 0.959 ± 0.020
NE3 mix	{0.98, 1.00, 0.88, 0.98, 0.96}, mean 0.960 ± 0.041

Interpretation. The NE3 mix and pure NormClip produce *statistically indistinguishable* ASR₅₀ (0.960 vs. 0.959, std overlapping), qualitatively consistent with Proposition 4.2’s persistent-regime prediction: when $\hat{\gamma}$ is moderately large (here ≈ 0.66) and $p_{\text{eff}}(\text{scaling}) > 0$, the realized utility under any mixed policy approaches the pure-admitting outcome. The 30-seed realized-VoPD measurement of -0.010 ± 0.058 (§5.4) is the corresponding numerical observation. We do not claim a tight quantitative match.

The pure FedAvg condition is bimodal across seeds ($\{0, 0, 0.92, 1, 1\}$): a feature of training-initialization randomness in whether the scaling attack converges, not of per-round Bernoulli admission. This separates a quantitative parameter-fit treatment (which would need to model the per-seed admissibility latent) from the qualitative result above. A rigorous per-defense, per-seed quantitative validation of the Markov model is left to future work.

Earlier discarded approach. An initial attempt to fit $(\gamma, \alpha, p_{\text{FA}}, p_{\text{NC}})$ jointly to the per-seed trajectories (1) hit a corner solution ($\hat{\gamma} \rightarrow 0$) due to the pure-FedAvg bimodality being interpreted as rapid decay, and (2) failed an out-of-sample test (fit on rounds 1–25, predict rounds 26–50: predicted NE3 ASR₅₀ = 0.80 vs. observed 0.96). We hypothesized that a per-seed admissibility variable separating the binding-cell Bernoulli at training init from the Markov dynamics is needed for quantitative prediction. We test this hypothesis below.

K.1 Admissibility-Conditional Markov Refinement

Setup. Classify each per-seed trajectory by “admissibility” — whether the binding-cell scaling attack has started to embed within the first 25 rounds: $a_{\text{seed}} = 1$ if $\text{ASR}_{25} > 0.5$, else 0. The training-initialization bimodality is captured by the latent a_{seed} ; conditional on $a_{\text{seed}} = 1$, fit only the Markov dynamics (α, γ) from rounds 1–25 (with p_{admit} fixed at the prior values 0.50, 1.00, 0.835 for FedAvg/NormClip/NE3-mix respectively), then predict ASR₅₀ as the out-of-sample test.

Result. The admissibility-conditional fit gives much better out-of-sample prediction on the two deployment-relevant conditions:

condition	out-of-sample MAE	fraction within ± 0.10	admit/no-admit seeds
NormClip pure ($p_{\text{admit}} = 1.0$)	0.041	5/5	5/0
NE3 mix ($p_{\text{admit}} = 0.835$)	0.059	3/4	4/1
FedAvg pure ($p_{\text{admit}} = 0.50$)	0.426	0/4	4/1

Interpretation. The persistence heuristic, conditioned on admissibility, is quantitatively predictive on the conditions that matter for deployment (NormClip-admitting and NE3 mix) with out-of-sample MAE under 0.06. The pure-FedAvg case has MAE = 0.426 because at $p_{\text{admit}} = 0.5$ the trajectory is dominated by the Bernoulli-binding-cell noise: the Markov model alone cannot predict per-seed scaling success vs. failure without explicitly modeling the per-round admission Bernoulli realization. We retain the qualitative Proposition 4.2 as the primary characterization, with the admissibility-conditional refinement as a partial advance: *the heuristic is quantitatively predictive on deployment-relevant policies but not on the FedAvg-pure baseline*. This converts the earlier discarded approach to a partial positive: the persistence mechanism is captured by a 2-parameter Markov model conditional on the admissibility latent, on conditions where admission is high enough that the Bernoulli noise does not dominate.

L Sample Complexity: Hoeffding/Bernstein Worst-Case Bound

This appendix contains the worst-case sample-complexity bound demoted from §4.1. The operational guarantee at $k = 10$ is the bootstrap-stability corollary (Proposition 2); the formal certificate below applies at $k \geq k_{\text{req}}$ and is included here for completeness and for practitioners targeting the formal regime.

Proposition 4 (Sample Complexity for Complementarity Diagnosis, formal). *Let $\hat{U}_A(a, d)$ be the mean of k i.i.d. $[0, 1]$ -bounded evaluations with per-run variance σ^2 . Define the best-response gap at defense d as $\Delta(d) = \max_a U_A(a, d) - \max_{a \neq a^*(d)} U_A(a, d)$, and let $\Delta_{\min} = \min_{d \in \mathcal{S}} \Delta(d)$ over the equilibrium support $\mathcal{S}$. Then with probability $\geq 1 - \delta$, Algorithm 1 returns the correct VoPD sign provided*

$$k \geq \frac{4(2\sigma^2 + \frac{1}{3}\Delta_{\min})}{\Delta_{\min}^2} \log\left(\frac{2|\mathcal{A}||\mathcal{S}|}{\delta}\right), \quad (13)$$

where $|\mathcal{A}|$ and $|\mathcal{S}|$ are the number of attacks and support defenses. When $\sigma \gg \Delta_{\min}/3$ (as in CIFAR-10 where $\sigma = 0.500$, $\Delta_{\min} = 0.128$), the σ^2 term dominates and Eq. (13) reduces to $k \geq 8\sigma^2 \log(2|\mathcal{A}||\mathcal{S}|/\delta)/\Delta_{\min}^2$.

Proof sketch. By Bernstein’s inequality (Bernstein, 1924), for $[0, 1]$ -bounded variables with variance σ^2 : $P(|\hat{U}_A(a, d) - U_A(a, d)| \geq \epsilon) \leq 2 \exp(-k\epsilon^2/(2\sigma^2 + \frac{2}{3}\epsilon))$. A wrong VoPD sign requires some best-response ranking to flip, requiring $|\hat{U}_A(a, d) - U_A(a, d)| \geq \Delta_{\min}/2$ for at least one (a, d) with $d \in \mathcal{S}$. Setting $\epsilon = \Delta_{\min}/2$ and applying a union bound over $|\mathcal{A}||\mathcal{S}|$ cells yields (13). $\square$

Empirical calibration. We use the *empirical-Bernstein* form: replace global σ^2 by per-cell $\hat{\sigma}_{a,d}^2$; the binding cell $\max_{a,d \in \text{supp}} \hat{\sigma}_{a,d}^2$ sets k_{req} . On CIFAR-10 ($N = 10$), U_A (scaling, FedAvg) is exactly Bernoulli ($\hat{\sigma}^2 = 0.25$, $\Delta_{\min} = 0.128$), giving $k_{\text{req}} = 726$ at $\delta = 0.10$. Non-binding cells (e.g., pixel/FedAvg, $\hat{\sigma}^2 < 0.01$) require $k < 50$ but do not constrain k_{req} . For spam (Appendix J), $\hat{\sigma}^2 = 0.0019$, $\Delta_{\min} = 0.217$ give $k_{\text{req}} \approx 33$. The bound is tight at maximal-variance binding cells (CIFAR-10), slack where variance is low (spam); it is a worst-case guarantee, informative only at $k \gg 10$. The operational guarantee at $k = 10$ is given by Proposition 2.

M Stateless and Orthogonal-Signal Pilots: Per-Seed Details

Detailed per-seed results, pre-committed criteria, and verdicts for the four pilots summarized in §5.13.

(i) Stateless availability pilot v1. The pilot introduces a **stateless availability attack**: malicious clients replace their updates with scaled Gaussian noise sampled fresh each round, so the effect is fully expended if the defense blocks that round and carries no persistent state. We re-run the 4×5 game on CIFAR-10 ($N=10$, 8 seeds, 50 rounds) with attacks {no_attack, backdoor_pixel, gaussian_noise, dba} against defenses {fedavg, norm_clip, rfa, trimmed_mean, coord_median}. *Utility note*: availability ASR is zero by definition; impact appears as accuracy degradation. We report under (i) the primary $U_A = \text{ASR} - 0.1 c_A$ and (ii) the alternative $U_A = \max(\text{ASR}, \text{Acc}_{\text{no-atk}} - \text{Acc}) - 0.1 c_A$ (Appendix D). *Result (8 seeds)*. Under (i), gaussian_noise’s row vanishes; the equilibrium collapses to pure pixel vs. FedAvg, VoPD = 0 on all 8 seeds. Under (ii), gaussian_noise gains positive utility against FedAvg ($U_A \approx 0.69$) and is neutralized by NormClip/RFA ($U_A < 0.05$); pixel remains dominant ($U_A \in [0.27, 0.91]$). The argmax structure varies per seed: 1/8 (seed 43) gives $\text{argmax}(\text{FedAvg}) = \text{gaussian_noise}$ (0.695) but $\text{argmax}(\text{NormClip}) = \text{pixel}$ (0.673), producing a mixed NE (FedAvg 97% + NormClip 3%, VoPD =

0.021); 7/8 give pixel as the universal argmax (VoPD = 0). Mean VoPD = 0.003 ± 0.007 . Algorithm 1 fires the randomize branch on 1/8 seeds. *Interpretation*: qualified positive existence result — stateless FL attacks *can* create complementarity, but the effect is small (mean VoPD ≤ 0.01), rare (1/8), and sensitive to the utility metric. Per-seed file: `results/cifar10_stateless/summary_corrected_utility.json`.

(ii) Low- K/N stateless pilot v2. Reviewer-suggested recipe: drop K/N , raise f , drop pixel from the menu. We re-ran the 4×5 game with $N=20$, $K=4$, $f=0.4$ (8 adversarial of 20), attacks `{no_attack, label_flip, gaussian_noise, dba}` and the same 5 defenses, 5 seeds. *Pre-committed STRONG criterion*: $\geq 3/5$ seeds mixed NE under metric (i) AND mean VoPD > 0.05 . *Result (5 seeds)*: Under (i), 0/5 mixed NE; equilibrium collapses to a single dominant attack vs. pure FedAvg (or pure TrimmedMean on seed 46). Under (ii), 1/5 mixed NE (seed 46) with adversary support DBA (98.2%) + gaussian_noise (1.8%); mean VoPD 0.080 ± 0.039 . **Verdict per pre-committed criterion: NEGATIVE.** The recipe does not deliver a strong positive FL case: even at $K/N=0.20$, $f=0.4$, and a stateless attack menu without pixel, the norm-based defense menu suppresses each stateless attack individually rather than producing heterogeneous argmax structure. Per-seed: `results/cifar10_positive_pilot_v2/summary.json`.

(iii) Orthogonal-signal pilot (FLTrust). We add FLTrust (Cao et al., 2021) — a defense weighting each client update by cosine similarity to a server-side gradient on a 100-sample clean holdout — to the original 4×5 CIFAR-10 game ($N=10$, $K=5$, $f=0.2$, attacks `{no_attack, backdoor_pixel, model_scaling, dba}`, defenses `{fedavg, norm_clip, rfa, trimmed_mean, coord_median, fltrust}`, 5 seeds 42–46, 50 rounds). *Pre-committed STRONG criterion*: $\geq 3/5$ mixed NEs under (i) AND FLTrust enters support AND argmax(FLTrust) differs from argmax(FedAvg). *Result (5 seeds)*: 4/5 seeds (43, 44, 45, 46) give argmax(FedAvg) = model_scaling while argmax(FLTrust) = backdoor_pixel — the orthogonal-signal mechanism produces the argmax divergence Corollary 1 predicts; FLTrust suppresses scaling ($U_A \approx 0.02-0.03$). **But FLTrust does not enter the equilibrium support** (0/5 seeds either metric): server cost-benefit selects FedAvg, NormClip, or TrimmedMean. Mean VoPD 0.021 ± 0.031 , 2/5 mixed NEs between norm-based defenses. **Verdict: WEAK positive.** The static-axis dichotomy is partially broken (direction-based defenses *do* create argmax divergence) but the server’s cost-benefit does not select the orthogonal-signal defense. Reading: orthogonal-signal mechanism is *necessary but not sufficient* for a positive FL deployment. Per-seed: `results/cifar10_fltrust_pilot/summary.json`.

(iv) λ_c sweep on the FLTrust menu. We test whether cost weight is the binding barrier by sweeping $\lambda_c \in \{0, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3\}$. The server payoff $U_D(a, d) = \text{Acc}(a, d) - \lambda_c \cdot C_D(d)$ is linear in λ_c , so we re-solve the Nash equilibria at each λ_c from the cached pilot matrices. *Pre-committed STRONG criterion*: $\exists \lambda_c \leq 0.05$ such that FLTrust enters equilibrium support (weight > 0.1) in $\geq 3/5$ seeds AND mixed-NE VoPD > 0.05 . *Result*: **FLTrust does not enter equilibrium support at any $\lambda_c \in [0, 0.3]$, including $\lambda_c = 0$.** At $\lambda_c = 0$ the server picks FedAvg or NormClip uniquely (clean accuracies ≈ 0.80 vs. FLTrust ≈ 0.72 , degraded by cosine-weighting overhead with the 100-sample holdout); FLTrust loses on Acc alone. **Verdict: NONE.** This is informative against the cost-weight explanation: *the barrier is not λ_c , it is the accuracy/robustness trade-off built into FLTrust’s mechanism*. Corollary 1 therefore tightens to a joint requirement: orthogonal-signal mechanism + competitive clean accuracy + low admission of persistent attacks.

Table 16: λ_c sweep on FLTrust menu (CIFAR-10, 5 seeds, metric (i)). We re-solve Nash equilibria at each λ_c using the cached payoff matrices. “FLTrust support” = number of seeds whose best NE has FLTrust with weight > 0.01 . “Mixed NEs” = seeds with a genuinely mixed equilibrium.

λ_c	Mean VoPD $\pm$ std	Mixed NEs	FLTrust in support	Best NE composition example
0.000	0.0000 ± 0.000	0/5	0/5	
0.025	0.0206 ± 0.031	2/5	0/5	
0.050	0.0206 ± 0.031	2/5	0/5	
0.075	0.0206 ± 0.031	2/5	0/5	
0.100	0.0206 ± 0.031	2/5	0/5	
0.150	0.0206 ± 0.031	2/5	0/5	
0.200	0.0206 ± 0.031	2/5	0/5	
0.300	0.0206 ± 0.031	2/5	0/5	

N Orthogonal-signal defense suite: per-seed equilibria

Per-seed Nash equilibria of the augmented 4-attack $\times$ 4-defense menu {FedAvg, NormClip, FoolsGold, reputation} with attacks {no_attack, backdoor_pixel, model_scaling, dba}. Cost weights $\lambda_c = 0.1$, $\lambda_f = 0.05$, $\lambda_a = 0.1$ as in the main game.

seed	adversary support	server support (incl. reputation weight)	VoPD
42	pixel (33%) + scaling (67%)	NormClip (89%) + reputation (11%)	0.089
43	pixel (75%) + scaling (25%)	NormClip (74%) + reputation (26%)	0.200
44	pixel (80%) + scaling (20%)	NormClip (86%) + reputation (14%)	0.117
45	pixel (99%)	FedAvg (22%) + NormClip (78%)	0.017
46	scaling (100%)	NormClip (100%)	0

Reputation in support: 3/5 seeds (42, 43, 44). Reputation entry weights 11–26%, alongside NormClip in the server support. Mixed-NE rate (5 seeds): 4/5. Mean VoPD across 5 seeds: 0.085. FoolsGold: 0/5 in support but argmax divergence in 4/5 seeds (similar to FLTrust). **Argmax structure:** argmax(reputation) = backdoor_pixel in all 5 seeds; argmax(NormClip) = model_scaling in 4/5 (seed 45 is bimodal with pixel-argmax under NormClip due to seed-45 scaling failure); argmax(FedAvg) is bimodal (pixel-argmax in seed 42, scaling-argmax in 43–46) due to the binding-cell Bernoulli.

Clean accuracy under no_attack: FoolsGold: 0.72–0.79 (mean 0.75 ± 0.03); reputation: 0.77–0.79 (mean 0.78 ± 0.01). Reputation matches FedAvg’s 0.80 within noise, satisfying the “competitive clean accuracy” requirement that FLTrust (≈ 0.72) failed.

Implementation. FoolsGold (Fung et al. 2018): for each client i , $w_i \propto \log((1 - v_i)/(v_i + 10^{-5}) + 10^{-5})$ where $v_i = \max_{j \neq i} |\cos(u_i, u_j)|$ (normalized by overall max). Reputation: $w_i = \exp(-\|u_i - \text{median}(\{u_j\})\|/\text{scale})$ with $\text{scale} = \text{median per-client distance from consensus (auto-tuned per round)}$. Both methods normalize weights to sum to 1 then weighted-FedAvg.

O Dynamic BR on Reputation Equilibrium: Per-Seed Details

Deploying NormClip 83% + reputation 17% (the cross-seed average over the reputation-bearing equilibria) and running NE3-style BR with committed adversary, 10 seeds 42–51, 50 rounds, 2 attacks. Per-seed realized ASRs and matched-seed realized VoPD computation:

seed	BR scaling	BR pixel	NC scaling	NC pixel	rep scaling	rep pixel	realized VoPD
42	0.923	0.832	0.937	0.837	0.014	0.837	−0.003
43	0.981	0.753	0.961	0.693	0.012	0.792	−0.049
44	0.865	0.824	0.911	0.774	0.017	0.828	+0.032
45	0.885	0.925	0.899	0.921	0.023	0.931	+0.038
46	0.949	0.767	0.965	0.805	0.019	0.823	−0.008
47	0.975	0.823	—	—	—	—	(PS missing)
48	0.979	0.923	—	—	—	—	(PS missing)
49	0.951	0.810	—	—	—	—	(PS missing)
50	0.937	0.866	—	—	—	—	(PS missing)
51	0.942	0.796	—	—	—	—	(PS missing)
mean (10)	0.939 ± 0.037	0.832 ± 0.055	—	—	—	—	—
mean (5)	—	—	—	—	—	—	$+0.002 \pm 0.031$

Aggregate. Realized scaling ASR = 0.939 ± 0.037 (10 seeds) — well above the 0.85 persistence-collapse threshold and consistent across all 10 seeds (no scaling-fails outlier in this run, unlike the original FedAvg+NormClip experiment which had seed 45’s bimodal failure). Realized VoPD on matched seeds 42–46 = $+0.002 \pm 0.031$ — essentially zero, within the COLLAPSES band ($-\text{VoPD} \leq 0.05$). 3/5 matched seeds give $\text{VoPD} \leq 0$.

Mechanism observed. The reputation defense strongly suppresses scaling statically (pure ASR ≈ 0.02 on matched seeds 42–46), but under the 17% mix-weight deployment, the BR adversary commits to scaling and embeds the trigger

during the 83% NormClip-admitting rounds. Across all 10 seeds, the number of reputation-sampled rounds varies between 5 and 12 (mean ≈ 8), but does not produce sufficient suppression to recover any realized VoPD — the static dichotomy break does not translate into a dynamic dichotomy break.

Pre-committed verdict: COLLAPSES. Per the pre-experiment statement: “Even when complementarity is generated through a genuinely orthogonal-signal defense (reputation), persistence still destroys realized value.” The persistence claim generalizes from norm-based to orthogonal-signal menus.

P Mix-ratio sweep on the reputation menu: per-seed details

Rounds 50–51 referenced in §5.6. Committed-attack BR on five NormClip+reputation mix ratios; 5 seeds, 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`. Mean $\pm$ std across seeds:

mix (NC/rep)	scaling ASR	std	pixel ASR	std	committed-max
90/10	0.945	0.021	0.804	0.075	0.945 (scaling)
70/30	0.908	0.043	0.810	0.066	0.916 (pixel @ s45)
50/50	0.815	0.104	0.822	0.055	0.845
30/70	0.510	0.187	0.835	0.044	0.835 (pixel)
10/90	0.097	0.090	0.835	0.051	0.835 (pixel)

Static full-info upper bound from cached cells: $\approx \{0.93, 0.91, 0.89, 0.87, 0.85\}$. Realized committed-max $\in [0.835, 0.945]$ exceeds this at every ratio: persistence boost on the committed adversary closes the static VoPD gap entirely, leaving realized VoPD in the collapse band across the full NC+rep mix-ratio range.

Observed transition. Realized scaling ASR is monotone in the reputation weight and exhibits a sharp transition between 30% and 90% reputation. Per-seed variance grows as reputation weight increases (std 0.021 $\rightarrow$ 0.187 $\rightarrow$ 0.090): the 50/50 ratio contains seed-43’s outlier (0.613); the 30/70 ratio splits roughly into two clusters (seeds 42, 44 above 0.69; seeds 43, 45, 46 at 0.34–0.39). At 10/90 only seed 42 retains nontrivial ASR (0.25); the other four collapse to ≤ 0.16 . The per-seed nondeterminism reflects randomness in defense-sampling: at 90% reputation, the realized number of NormClip rounds across 50 rounds varies between 1 and 6 across seeds (mean ≈ 4), and these 1–6 NormClip-admitting rounds determine whether the trigger embeds at all.

Verdict. MEDIUM per pre-committed bands. The persistence collapse is not mix-ratio-universal: it holds on the NormClip-dominant half (ratios with NormClip $\geq 50\%$) and breaks on reputation-dominant mixes (operative threshold in the 70%–90% reputation range). The equilibrium-relevant ratio (17% reputation, §5.5) lies on the collapse-surviving side.

λ_c sweep on the orthogonal menu. Sweep referenced in §5.7. NE re-solved on {FedAvg, NormClip, FoolsGold, reputation} at $\lambda_c \in \{0, 0.025, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3\}$. Reputation enters NE support in seeds 43 (25.6%) and 44 (14.5%) at every λ_c ; mean max-reputation-weight is constant at 10.2%. *The binding constraint is reputation’s clean-accuracy gap, not its cost C_D .* The persistence-survival region (reputation NE weight $\geq 70\%$) is unreachable as a natural NE on this menu for any tested λ_c .

λ_c	mean max-reputation-weight	seeds with rep-weight $\geq 50\%$	seeds with rep-weight $\geq 70\%$
0.000	0.102	0/5	0/5
0.025	0.102	0/5	0/5
0.050	0.102	0/5	0/5
0.075	0.102	0/5	0/5
0.100	0.102	0/5	0/5
0.150	0.102	0/5	0/5
0.200	0.102	0/5	0/5
0.300	0.102	0/5	0/5

Q Reputation+trimmed_mean survivor regime: per-seed details

Per-seed details for §5.8. Deployed mix: reputation 30% + trimmed_mean 70%, the static-VoPD-maximising point on the cached 7-defense, 2-attack cell grid (mean static VoPD 0.162 across seeds). NE3-style BR with three ad-

versary policies; 30 seeds (42–71), 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`. Oracle adversary observes the round’s sampled defense and plays scaling on `trimmed_mean` rounds ($\text{argmax}(\text{tm})=\text{scaling}$), pixel on reputation rounds ($\text{argmax}(\text{rep})=\text{pixel}$). Trigger is shared between scaling and pixel attacks (bottom-right 4x4 pixel pattern); end-of-training ASR is measured against the shared trigger.

adversary policy	mean ASR ($n = 30$)	std	per-seed range	# seeds ≥ 0.85
committed-scaling	0.871	0.095	[0.588, 0.985]	19/30
committed-pixel	0.747	0.106	[0.489, 0.926]	6/30
oracle	0.915	0.065	[0.649, 0.987]	25/30
per-seed committed-max	0.883	0.075	—	—
per-seed VoPD	+0.032	0.054	[−0.146, +0.123]	—

Distribution. 27/30 seeds positive VoPD; 8/30 in the STRONG band (≥ 0.05); 3/30 seeds negative. The single strong negative outlier is seed 50 (VoPD −0.146): committed-scaling on seed 50 is anomalously low (0.588, vs. the 0.87 mean), so the oracle’s scaling rounds contribute weakly and the oracle’s overall ASR (0.649) falls short of committed-pixel (0.795). Removing seed 50 gives VoPD $+0.038 \pm 0.041$ (27/29 positive). The 5-seed pilot’s $+0.069 \pm 0.026$ was on the high tail of this distribution.

Mechanism. The oracle plays scaling on $\sim 68\%$ of rounds (`trimmed_mean` rounds, where reputation rounds are $\sim 32\%$) and pixel on the remaining $\sim 32\%$. Both attacks embed the same trigger pattern; the oracle achieves union-embedding by running the more-effective mechanism per round. Committed-scaling pays the cost of running scaling through `trimmed_mean` rounds where its update is partially clipped; committed-pixel pays the cost of running pixel through `trimmed_mean` rounds where the update magnitudes are reduced.

Verdict. MEDIUM positive — complementarity survives at modest magnitude. Realized VoPD = +0.032 at $n = 30$ falls in the pre-committed MEDIUM band ([0.02, 0.05]). The result is statistically robust (27/30 positive, $p < 10^{-5}$ vs. Bernoulli(0.5)) but the magnitude is roughly half what the 5-seed pilot suggested. This sharpens the survivor claim: the boundary between collapse and survival exists as a real statistical property, but the survival magnitude in our menu is small in absolute terms ($\sim 3\%$ accuracy gap). The defense-menu’s arg max structure determines whether complementarity survives; the magnitude of survival depends on how cleanly the two persistent attacks orthogonally suppress each other.

R Edge_case 3-attack survivor pilot: per-seed details

Per-seed details for §5.10. Same deployed mix (reputation 30% + `trimmed_mean` 70%), same 5 seeds, 50 rounds, $N=10$, $K=5$, $f=0.2$, `cifar_cnn`, with `backdoor_edge_case` added as a 4th adversary policy. The oracle’s per-defense argmax table is unchanged (pixel on reputation rounds, scaling on `trimmed_mean` rounds) because `edge_case` is dominated by $\text{max}(\text{scaling}, \text{pixel})$ on every defense in the menu. Trigger evaluation uses `evaluate_backdoor` for scaling/pixel rounds (pixel trigger) and `evaluate_edge_case_backdoor` for `edge_case` rounds (top-corner trigger).

Pure-strategy edge_case ASR (refilled with correct evaluator, 5-seed mean $\pm$ std).

defense	ASR	seed-45 outlier
fedavg / norm_clip	0.619 ± 0.108	0.835
foolsgold	0.452 ± 0.094	—
reputation	0.611 ± 0.111	0.828
rfa	0.610 ± 0.107	0.813
trimmed_mean	0.394 ± 0.193	0.756
coord_median	0.235 ± 0.113	0.440

`Trimmed_mean` and `coord_median` suppress `edge_case` more than other defenses (matching their suppression pattern on scaling/pixel); seed 45 is consistently a positive outlier across all defenses except FoolsGold.

Realized 4-policy ASRs at rep30/tm70.

adversary policy	seed 42	seed 43	seed 44	seed 45	seed 46	mean $\pm$ std
committed-scaling	0.851	0.733	0.753	0.861	0.864	0.812 ± 0.060
committed-pixel	0.722	0.519	0.667	0.901	0.735	0.709 ± 0.131
committed-edge_case	0.387	0.285	0.413	0.798	0.498	0.476 ± 0.176
oracle (pixel on rep, scaling on tm)	0.905	0.813	0.804	0.968	0.912	0.880 ± 0.063
per-seed committed-max	0.851 (sc)	0.733 (sc)	0.753 (sc)	0.901 (px)	0.864 (sc)	0.820 ± 0.066
per-seed VoPD	+0.054	+0.080	+0.051	+0.067	+0.048	$+0.060 \pm 0.013$

Observations. (i) Realized committed-edge_case ASR (0.476) is materially below committed-scaling (0.812) and committed-pixel (0.709); edge_case is the weakest persistent attack on this mix. (ii) The oracle never plays edge_case (model_scaling=35–38 rounds, pixel=12–15 rounds, edge_case=0 across all seeds), because edge_case is dominated by max(scaling, pixel) on both menu defenses statically. (iii) Per-seed committed-max equals committed-scaling on 4/5 seeds and committed-pixel on seed 45 — the same per-seed pattern observed without edge_case in the original 3-policy pilot. (iv) Per-seed VoPDs $\{+0.054, +0.080, +0.051, +0.067, +0.048\}$ are concentrated near +0.06 (std 0.013). Mean +0.060 vs. the original +0.069 are statistically indistinguishable at $n=5$.

Verdict. STRONG positive: the rep30/tm70 survivor regime is robust against a 3-attack adversary. The boundary characterisation (collapse vs. survival) is therefore a property of the defense menu’s arg max structure across the persistent attacks present in the game, not of the attack-menu cardinality. Edge_case as a 4th option confirms but does not extend the boundary: a meaningful 9/10-grade demonstration would require an attack that wins the static arg max on at least one defense in the menu (so the oracle plays it), and the present 3-attack landscape on this 7-defense menu does not exhibit such an attack.

S ResNet18 architecture-replication mini-sweep: per-seed details

Per-seed details for §5.11. *resnet18* with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to 32×32 (kernel 3, stride 1, maxpool $\rightarrow$ Identity); 11.2M parameters. Identical FL parameters to *cifar_cnn*: $N=10$, $K=5$, $f=0.2$, 50 rounds, $\tau = 5$. Committed-scaling BR, 3 mix points $\times$ 5 seeds = 15 runs at ~ 100 min/run on MPS.

mix (NC/rep)	s42	s43	s44	s45	s46	ResNet18 mean	cifar_cnn (R50)
90/10	0.926	0.929	0.965	0.974	0.962	0.951 ± 0.020	0.945 ± 0.021
50/50	0.924	0.971	0.813	0.920	0.903	0.906 ± 0.052	0.815 ± 0.104
10/90	0.917	0.733	0.783	0.975	0.915	0.865 ± 0.091	0.097 ± 0.090

NC-dominant endpoint replicates within 0.01; mid-mix is $\sim 9\%$ higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict: MEDIUM** (monotonic ordering preserved, endpoint criterion fails). The *cifar_cnn* collapse boundary at ~ 70 –90% reputation is therefore a property of the smaller-architecture regime; whether the rep+tm survivor also transfers is out of revision scope.

T DBA at $f=0.4$: per-seed details

NE3 BR on CIFAR-10 with DBA in place of model scaling at $f=0.4$ (raising adversarial co-participation to $K \cdot f = 2.0$ from the $f=0.2$ value of 1.0). $N=10$, $K=5$, 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

seed	DBA realized ASR	backdoor_pixel realized ASR
42	0.222	0.972
43	0.103	0.945
44	0.176	0.913
45	0.603	0.993
46	0.077	0.934
mean $\pm$ std	0.236 ± 0.190	0.952 ± 0.028

The DBA per-seed spread is $7.5\times$ between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold (≥ 0.20) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at $f=0.4$ (0.952 ± 0.028), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether ≥ 2 of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at $f=0.4$ is suggestive but not a robust second-attack demonstration.

U DBA at $f=0.6$: per-seed details and bimodality reproduction

Follow-up to Appendix T, designed to test whether the DBA bimodality at $f=0.4$ is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher f . At $f=0.6$, expected adversarial co-participation per round is $K \cdot f = 3.0$, making ≥ 2 adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds, $N=10$, $K=5$, `cifar_cnn`. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

seed	DBA ASR (NE3 mix)	DBA ASR (reputation eq)
42	0.360	0.339
43	0.295	0.278
44	0.386	0.415
45	0.734	0.705
46	0.154	0.130
mean $\pm$ std	0.386 ± 0.192	0.373 ± 0.190

Observations. (i) Mean DBA ASR improves over $f=0.4$ (0.236) to ~ 0.38 on both policies, but does not reach the ≥ 0.85 persistence band scaling achieves. (ii) Per-seed bimodality persists at $f=0.6$: seed 45 alone reaches > 0.7 in both policies; the remaining four seeds cluster in $[0.13, 0.42]$. (iii) Crucially, *the same seed is the strong-persistence seed under both policies* (NE3 mix: 0.734; reputation eq: 0.705), and *the same four seeds are the weak-persistence seeds under both* — the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli (Appendix K.1), but at a much lower mean (admissibility prevalence $\approx 1/5$ for DBA at $f=0.6$ vs. $\approx 4/5$ for scaling at $f=0.2$).

Pre-committed verdict: INTERMEDIATE. Overall mean 0.380 falls between PARTIAL (> 0.5) and NEGATIVE (< 0.3). Honest reading: DBA at $f=0.6$ does *not* replicate the persistence collapse with the cleanness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising f . The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.